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Cloud Computing Lab

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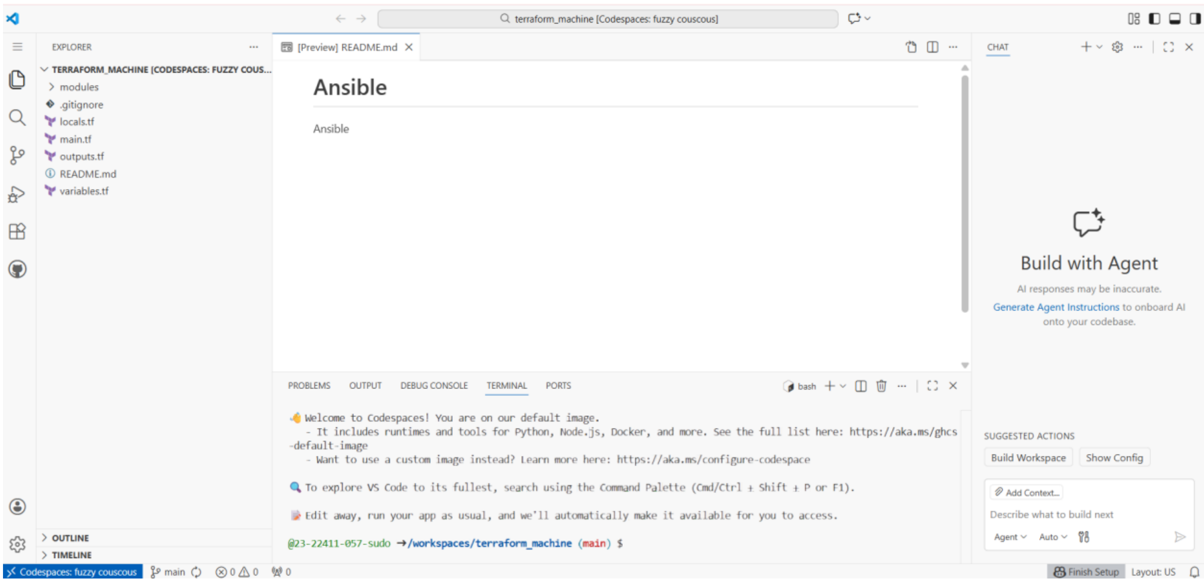
Section: 5-B

LAB NO 14

Task 0 – Lab Setup (Codespace & GH CLI)

- task0_codespace_open.png



- task0_env_check.png

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ gh auth status
github.com
  ✓ Logged in to github.com account 23-22411-057-sudo (GITHUB_TOKEN)
  - Active account: true
  - Git operations protocol: https
  - Token: ghu_************************************
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ aws --version
aws-cli/2.32.31 Python/3.13.11 Linux/6.8.0-1030-azure exe/x86_64.ubuntu.24
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform --version
Terraform v1.14.3
on linux_amd64
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/codespace/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/codespace/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Nov  6 2025, 13:44:16) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.6
  libyaml = True
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

- task0_aws_config.png

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ aws sts get-caller-identity
{
    "UserId": "AIDAVHLXGUTLOH3DDK36F",
    "Account": "359416636630",
    "Arn": "arn:aws:iam::359416636630:user/Admin"
}
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

Task 1 – Generate ssh key and Initial Terraform apply

1. Check SSH directory & generate SSH key pair if not already present:

task1_ssh_keygen_before.png — ls ~/.ssh before key generation.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ ls ~/.ssh
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -N ""
```

task1_ssh_keygen.png — output of ssh-keygen command.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -N ""
Generating public/private ed25519 key pair.
Your identification has been saved in /home/codespace/.ssh/id_ed25519
Your public key has been saved in /home/codespace/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:v+snEXeH/01bRgNoyZRdP1uIcqkVEOIDj3Aw6hSZULM codespace@codespaces-94f032
The key's randomart image is:
+--[ED25519 256]--+
|.o+o+.o . =+* .. |
|  o= + = . * * ..|
|  E   . + o = ooo|
| o       ..=. o.*|
|  .     S .o . =.|
|         ..     =|
|          ..   o=|
|          ...  .o|
|         .++     |
+----[SHA256]-----+
```

task1_ssh_keygen_after.png — ls -la ~/.ssh showing id_ed25519 and id_ed25519.pub.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ ls -la ~/.ssh
total 20
drwx------ 2 codespace codespace 4096 Jan  9 16:58 .
drwxr-x--- 1 codespace codespace 4096 Jan  9 16:57 ..
-rw------- 1 codespace codespace  419 Jan  9 16:58 id_ed25519
-rw-r--r-- 1 codespace codespace  109 Jan  9 16:58 id_ed25519.pub
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

2. Create terraform.tfvars in the repo root:

task1_terraform_tfvars_created.png — touch terraform.tfvars and ls -la terraform.tfvars.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ cd /workspaces/terraform_machine
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ touch terraform.tfvars
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ ls -la terraform.tfvars
-rw-rw-rw- 1 codespace codespace 0 Jan  9 17:05 terraform.tfvars
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

task1_terraform_tfvars.png — content of terraform.tfvars.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS                    nano
 GNU nano 7.2                     terraform.tfvars *
vpc_cidr_block = "10.0.0.0/16"
subnet_cidr_block = "10.0.10.0/24"
availability_zone = "me-central-1a"
env_prefix = "dev"
instance_type = "t3.micro"
public_key = "~/.ssh/id_ed25519.pub"
private_key = "~/.ssh/id_ed25519"
```

3. Initialize Terraform:

task1_terraform_init.png — terraform init output.

```
● @23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform init
Initializing the backend...
Initializing modules...
- myapp-subnet in modules/subnet
- myapp-webserver in modules/webserver
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Finding latest version of hashicorp/http...
- Installing hashicorp/aws v6.28.0...
- Installed hashicorp/aws v6.28.0 (signed by HashiCorp)
- Installing hashicorp/http v3.5.0...
- Installed hashicorp/http v3.5.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
○ @23-22411-057-sudo →/workspaces/terraform_machine (main) $ 
```

4. **Apply Terraform** to create **2 EC2 instances** (as defined in the existing Terraform code):

Task1_terraform_apply_2_instances.png — terraform apply output showing 2 instances created.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform apply -auto-approve
    ]
aws_vpc.myapp_vpc: Creating...
module.myapp-webserver[0].aws_key_pair.ssh-key: Creating...
module.myapp-webserver[1].aws_key_pair.ssh-key: Creating...
module.myapp-webserver[1].aws_key_pair.ssh-key: Creation complete after 1s [id=dev-serverkey-1]
module.myapp-webserver[0].aws_key_pair.ssh-key: Creation complete after 1s [id=dev-serverkey-0]
aws_vpc.myapp_vpc: Still creating... [00m10s elapsed]
aws_vpc.myapp_vpc: Creation complete after 12s [id=vpc-0bf2e8de0352486a5]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creating...
module.myapp-subnet.aws_internet_gateway.myapp_igw: Creating...
module.myapp-webserver[0].aws_security_group.web_sg: Creating...
module.myapp-webserver[1].aws_security_group.web_sg: Creating...
module.myapp-subnet.aws_internet_gateway.myapp_igw: Creation complete after 1s [id=igw-01507100278956522]
module.myapp-subnet.aws_default_route_table.main_rt: Creating...
module.myapp-subnet.aws_default_route_table.main_rt: Creation complete after 1s [id=rtb-0cd5a79d0f7e93439]
module.myapp-webserver[0].aws_security_group.web_sg: Creation complete after 3s [id=sg-088adec1554a602fa]
module.myapp-webserver[1].aws_security_group.web_sg: Creation complete after 3s [id=sg-039ed491d98cbaaff]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Still creating... [00m10s elapsed]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creation complete after 11s [id=subnet-09842c89977df0aa4]
module.myapp-webserver[1].aws_instance.myapp-server: Creating...
module.myapp-webserver[0].aws_instance.myapp-server: Creating...
module.myapp-webserver[1].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Creation complete after 13s [id=i-0f5c904ff54acbb52]
module.myapp-webserver[1].aws_instance.myapp-server: Creation complete after 13s [id=i-09fbfd54e203f408d]

Apply complete! Resources: 10 added, 0 changed, 0 destroyed.

Outputs:

webserver_public_ips = [
  "3.29.33.227",
  "40.172.101.201",
]
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

5. Check outputs

task1_terraform_output_ips.png — terraform output showing instance IPs.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform output
webserver_public_ips = [
  "3.29.33.227",
  "40.172.101.201",
]
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

Task 2 – Static Ansible inventory with two EC2 instances

1. Install Ansible (core) using pipx:

task2_ansible_install.png — pipx install ansible-core and ansible --version output.

```
● @23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ pipx install ansible-core
  ⚠ Note: ansible was already on your PATH at /usr/bin/ansible
  ⚠ Note: ansible-config was already on your PATH at /usr/bin/ansible-config
  ⚠ Note: ansible-console was already on your PATH at /usr/bin/ansible-console
  ⚠ Note: ansible-doc was already on your PATH at /usr/bin/ansible-doc
  ⚠ Note: ansible-galaxy was already on your PATH at /usr/bin/ansible-galaxy
  ⚠ Note: ansible-inventory was already on your PATH at /usr/bin/ansible-inventory
  ⚠ Note: ansible-playbook was already on your PATH at /usr/bin/ansible-playbook
  ⚠ Note: ansible-pull was already on your PATH at /usr/bin/ansible-pull
  ⚠ Note: ansible-test was already on your PATH at /usr/bin/ansible-test
  ⚠ Note: ansible-vault was already on your PATH at /usr/bin/ansible-vault
    installed package ansible-core 2.20.1, installed using Python 3.12.1
    These apps are now globally available
      - ansible
      - ansible-config
      - ansible-console
      - ansible-doc
      - ansible-galaxy
      - ansible-inventory
      - ansible-playbook
      - ansible-pull
      - ansible-test
      - ansible-vault
  done! ✨ 🌟 ✨
● @23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ ansible --version
  ansible [core 2.16.3]
    config file = None
    configured module search path = ['/home/codespace/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
    ansible python module location = /usr/lib/python3/dist-packages/ansible
    ansible collection location = /home/codespace/.ansible/collections:/usr/share/ansible/collections
    executable location = /usr/bin/ansible
    python version = 3.12.3 (main, Nov  6 2025, 13:44:16) [GCC 13.3.0] (/usr/bin/python3)
    jinja version = 3.1.6
    libyaml = True
○ @23-22411-057-sudo ➜/workspaces/terraform_machine (main) $
```

2. **Obtain the two public IPs** of your EC2 instances:

task2_terraform_output_ips.png — terraform output listing at least 2 EC2 public IPs.

```
● @23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ terraform output
  webserver_public_ips = [
    "3.29.33.227",
    "40.172.101.201",
  ]
○ @23-22411-057-sudo ➜/workspaces/terraform_machine (main) $
```

3. **Create Ansible inventory file hosts**:

task2_hosts_created.png — confirmation that hosts file exists.

```
● @23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ touch hosts
● @23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ ls -la hosts
  -rw-rw-rw- 1 codespace codespace 0 Jan  9 17:22 hosts
○ @23-22411-057-sudo ➜/workspaces/terraform_machine (main) $
```

task2_hosts_initial.png — content of hosts with two IP lines.

```
GNU nano 7.2                                  hosts *
3.29.33.227 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519 ansible_ssh_common_args='-o StrictHo>
40.172.101.201 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519 ansible_ssh_common_args='-o Stric>
```

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ nano hosts
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ cat hosts
<public-ip-ec2-1> ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519
<public-ip-ec2-2> ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

4. **Test connectivity**:

task2_ansible_ping_initial.png — initial ansible all -i hosts -m ping output (success or failure).

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ ansible all -i hosts -m ping
[WARNING]: Platform linux on host 40.172.101.201 is using the discovered Python interpreter at /usr/bin/python3.9,
but future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.16/reference_appendices/interpreter_discovery.html for more information.
40.172.101.201 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Platform linux on host 3.29.33.227 is using the discovered Python interpreter at /usr/bin/python3.9, but
future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.16/reference_appendices/interpreter_discovery.html for more information.
3.29.33.227 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

Task 3 - Scale to three instances & group-based inventory

You will expand to 3 web servers via Terraform's count and restructure your inventory into groups.

1. **Update your Terraform module for webservers** in main.tf to use count = 3:

task3_main_tf_count_3.png — main.tf snippet showing count = 3.

```
 GNU nano 7.2                                              main.tf *
  tags = {
     Name = "${var.env_prefix}-vpc"
  }
}

module "myapp-subnet" {
  source = "./modules/subnet"
  vpc_id = aws_vpc.myapp_vpc.id
  subnet_cidr_block = var.subnet_cidr_block
  availability_zone = var.availability_zone
  env_prefix = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}

module "myapp-webserver" {
  source            = "./modules/webserver"
  env_prefix        = var.env_prefix
  instance_type     = var.instance_type
  availability_zone = var.availability_zone
  public_key        = var.public_key
  my_ip             = local.my_ip
  vpc_id            = aws_vpc.myapp_vpc.id
  subnet_id         = module.myapp-subnet.subnet.id

  # Loop count
  count = 3

  # Use count.index to differentiate instances
  instance_suffix = count.index
}
```

2. **Apply Terraform** to get 3 instances:

task3_terraform_apply_3_instances.png — terraform apply output creating 3 instances.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform apply -auto-approve
      + tags                    = {
          + "Name" = "dev-default-sg"
        }
      + tags_all                = {
          + "Name" = "dev-default-sg"
        }
      + vpc_id                  = "vpc-0bf2e8de0352486a5"
    }

Plan: 3 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  ~ webserver_public_ips = [
        # (1 unchanged element hidden)
        "40.172.101.201",
      + (known after apply),
    ]
module.myapp-webserver[2].aws_key_pair.ssh-key: Creating...
module.myapp-webserver[2].aws_security_group.web_sg: Creating...
module.myapp-webserver[2].aws_key_pair.ssh-key: Creation complete after 0s [id=dev-serverkey-2]
module.myapp-webserver[2].aws_security_group.web_sg: Creation complete after 3s [id=sg-0a018c86feda1c480]
module.myapp-webserver[2].aws_instance.myapp-server: Creating...
module.myapp-webserver[2].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver[2].aws_instance.myapp-server: Creation complete after 14s [id=i-0fb38402e2a321d85]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

webserver_public_ips = [
  "3.29.33.227",
  "40.172.101.201",
  "3.28.133.205",
]
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

3. **Check outputs**:

task3_terraform_output_3_ips.png — terraform output listing 3 public IPs.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform output
webserver_public_ips = [
  "3.29.33.227",
  "40.172.101.201",
  "3.28.133.205",
]
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

4. **Rewrite your hosts file** using group definitions:

task3_hosts_grouped.png — hosts showing [ec2] and [droplet] with three IPs.

```
GNU nano 7.2                                   hosts *
[ec2]
3.29.33.227
40.172.101.201

[ec2:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_ssh_common_args='-o StrictHostKeyChecking=no'

[droplet]
3.28.133.205

[droplet:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_ssh_common_args='-o StrictHostKeyChecking=no'
```

5. **Test group connectivity**:

task3_ansible_ec2_ping.png — output of ansible ec2 -i hosts -m ping.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ cat hosts
[ec2]
3.29.33.227
40.172.101.201

[ec2:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_ssh_common_args='-o StrictHostKeyChecking=no'

[droplet]
3.28.133.205

[droplet:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_ssh_common_args='-o StrictHostKeyChecking=no'

@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

6. **Test single host by IP**:

task3_ansible_single_ip_ping.png — ping output for one EC2 IP.

```
● @23-22411-057-sudo →/workspaces/terraform_machine (main) $ ansible ec2 -i hosts -m ping
[WARNING]: Platform linux on host 3.29.33.227 is using the discovered Python interpreter at /usr/bin/python3.9, but
future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.16/reference_appendices/interpreter_discovery.html for more information.
3.29.33.227 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Platform linux on host 40.172.101.201 is using the discovered Python interpreter at /usr/bin/python3.9,
but future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.16/reference_appendices/interpreter_discovery.html for more information.
40.172.101.201 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
○ @23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

```
● @23-22411-057-sudo →/workspaces/terraform_machine (main) $ ansible 3.29.33.227 -i hosts -m ping
[WARNING]: Platform linux on host 3.29.33.227 is using the discovered Python interpreter at /usr/bin/python3.9, but
future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.16/reference_appendices/interpreter_discovery.html for more information.
3.29.33.227 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
○ @23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

7. **Test droplet group**:

task3_ansible_droplet_ping.png — ping output for droplet group.

```
● @23-22411-057-sudo →/workspaces/terraform_machine (main) $ ansible droplet -i hosts -m ping
[WARNING]: Platform linux on host 3.28.133.205 is using the discovered Python interpreter at /usr/bin/python3.9, but
future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.16/reference_appendices/interpreter_discovery.html for more information.
3.28.133.205 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
```

8. **Test all hosts**:

task3_ansible_all_ping.png — ping output for all instances.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ ansible all -i hosts -m ping
[WARNING]: Platform linux on host 3.29.33.227 is using the discovered Python interpreter at /usr/bin/python3.9, but
future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.16/reference_appendices/interpreter_discovery.html for more information.
3.29.33.227 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Platform linux on host 40.172.101.201 is using the discovered Python interpreter at /usr/bin/python3.9,
but future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.16/reference_appendices/interpreter_discovery.html for more information.
40.172.101.201 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Platform linux on host 3.28.133.205 is using the discovered Python interpreter at /usr/bin/python3.9, but
future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.16/reference_appendices/interpreter_discovery.html for more information.
3.28.133.205 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

Task 4 – Global ansible.cfg & first nginx playbook

You will configure a **global Ansible configuration file**, then create a basic playbook for nginx.

1. Create global Ansible configuration:

task4_global_ansible_cfg.png — content of ~/.ansible.cfg.

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS vim

```
[default]
host_key_checking = False
interpreter_python = /usr/bin/python3

~
~
~
~
~
~
~
~
~
```

2. Remove ansible_ssh_common_args from hosts (delete from all groups).

task4_hosts_without_common_args.png — hosts after removal.

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                    vim
[ec2]
3.29.33.227
40.172.101.201

[ec2:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519

[droplet]
3.28.133.205

[droplet:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519

~
~
```

3. Confirm connectivity:

task4_ansible_ping_after_cfg.png — ping output showing success, no strict host checking warning.

```
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ ansible all -i hosts -m ping
[WARNING]: Host '3.29.33.227' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installat
ion of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/a
nsible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
3.29.33.227 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Host '3.28.133.205' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installa
tion of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/
ansible-core/2.20/reference_appendices/interpreter_discovery.html for more information.
3.28.133.205 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $
```

4. Create my-playbook.yaml:

task4_my_playbook_created.png — existence of my-playbook.yaml.

```
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ touch my-playbook.yaml
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ ls -la my-playbook.yaml
-rw-rw-rw- 1 codespace codespace 0 Jan 10 12:48 my-playbook.yaml
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $
```

task4_my_playbook_ec2.png — content of my-playbook.yaml with hosts: ec2.

```yaml
---
- name: Configure nginx web server
  hosts: ec2
  become: true
  tasks:
    - name: install nginx and update cache
      yum:
        name: nginx
        state: present
        update_cache: yes

    - name: start nginx server
      service:
        name: nginx
        state: started
```

5. **Run the playbook** on [ec2] group:

task4_ansible_play_ec2.png — playbook output for ec2 group.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml

PLAY [Configure nginx web server] ***********************************************************************************

TASK [Gathering Facts] **********************************************************************************************
[WARNING]: Host '3.29.33.227' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Pyt
hon interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendice
s/interpreter_discovery.html for more information.
ok: [3.29.33.227]

TASK [install nginx and update cache] *******************************************************************************
changed: [3.29.33.227]

TASK [start nginx server] *******************************************************************************************
changed: [3.29.33.227]

PLAY RECAP **********************************************************************************************************
3.29.33.227                : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

6. **Verify** nginx default page on [ec2] servers by visiting http://<public-ip-ec2>.

task4_nginx_browser_ec2.png — browser showing nginx default page for an ec2 instance.

7. **Change target to droplet**:

task4_my_playbook_droplet.png — playbook with hosts: droplet.

```
● @23-22411-057-sudo →/workspaces/terraform_machine (main) $ ansible droplet -i hosts -m ping
[WARNING]: Host '3.28.133.205' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Py
thon interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendic
es/interpreter_discovery.html for more information.
3.28.133.205 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
○ @23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

8. Re-run:

task4_ansible_play_droplet.png — output of playbook for droplet group.

```
---
- name: Configure nginx web server
  hosts: droplet
  become: true
  tasks:
    - name: install nginx and update cache
      yum:
        name: nginx
        state: present
        update_cache: yes

    - name: start nginx server
      service:
        name: nginx
        state: started

~
~
~
~
~
```

```
● @23-22411-057-sudo →/workspaces/terraform_machine (main) $ vim my-playbook.yaml
● @23-22411-057-sudo →/workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml

PLAY [Configure nginx web server] ************************************************************************************************

TASK [Gathering Facts] ***********************************************************************************************************
[WARNING]: Host '3.28.133.205' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another Py
thon interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendic
es/interpreter_discovery.html for more information.
ok: [3.28.133.205]

TASK [install nginx and update cache] ********************************************************************************************
changed: [3.28.133.205]

TASK [start nginx server] ********************************************************************************************************
changed: [3.28.133.205]

PLAY RECAP ***********************************************************************************************************************
3.28.133.205               : ok=3    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

9. Verify nginx default page on droplet:

task4_nginx_browser_droplet.png — browser showing nginx page on droplet.

Task 5 – Single nginx target group & HTTPS prerequisites

You will prepare **project-level Ansible configuration** and adjust nginx installation scope.

1. **Create project-level ansible.cfg** in repo root:

task5_project_ansible_cfg_created.png — project ansible.cfg presence.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ touch ansible.cfg
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ ls -la ansible.cfg
-rw-rw-rw- 1 codespace codespace 0 Jan 10 13:02 ansible.cfg
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

task5_project_ansible_cfg.png — project-level ansible.cfg content.

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                vim +

[defaults]
host_key_checking=False
interpreter_python = /usr/bin/python3

~
~
~
~
~
~
~
```

2. **Switch Terraform EC2 count back to 1**:

task5_main_tf_count_1.png — snippet showing count = 1.

```
resource "aws_vpc" "myapp_vpc" {
  cidr_block = var.vpc_cidr_block
  enable_dns_hostnames = true
  tags = {
     Name = "${var.env_prefix}-vpc"
  }
}

module "myapp-subnet" {
  source = "./modules/subnet"
  vpc_id = aws_vpc.myapp_vpc.id
  subnet_cidr_block = var.subnet_cidr_block
  availability_zone = var.availability_zone
  env_prefix = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}

module "myapp-webserver" {
  source            = "./modules/webserver"
  env_prefix        = var.env_prefix
  instance_type     = var.instance_type
  availability_zone = var.availability_zone
  public_key        = var.public_key
  my_ip             = local.my_ip
  vpc_id            = aws_vpc.myapp_vpc.id
  subnet_id         = module.myapp-subnet.subnet.id

  # Loop count
  count = 1

  # Use count.index to differentiate instances
  instance_suffix = count.index
}

-- INSERT --                                                           34,12         Bot
```

3. Apply:

task5_terraform_apply_one_instance.png — apply output with one instance.

```
● @23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform apply -auto-approve

  data.http.my_ip: Reading...
  data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]
  module.myapp-webserver[1].aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-1]
  aws_vpc.myapp_vpc: Refreshing state... [id=vpc-0bf2e8de0352486a5]
  module.myapp-webserver[2].aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-2]
  module.myapp-webserver[0].aws_key_pair.ssh-key: Refreshing state... [id=dev-serverkey-0]
  module.myapp-subnet.aws_internet_gateway.myapp_igw: Refreshing state... [id=igw-01507100278956522]
  module.myapp-subnet.aws_subnet.myapp_subnet_1: Refreshing state... [id=subnet-09842c89977df0aa4]
  module.myapp-webserver[0].aws_security_group.web_sg: Refreshing state... [id=sg-088adec1554a602fa]
  module.myapp-webserver[2].aws_security_group.web_sg: Refreshing state... [id=sg-0a018c86feda1c480]
  module.myapp-webserver[1].aws_security_group.web_sg: Refreshing state... [id=sg-039ed491d98cbaaff]
  module.myapp-subnet.aws_default_route_table.main_rt: Refreshing state... [id=rtb-0cd5a79d0f7e93439]
  module.myapp-webserver[1].aws_instance.myapp-server: Refreshing state... [id=i-09fbfd54e203f408d]
  module.myapp-webserver[2].aws_instance.myapp-server: Refreshing state... [id=i-0fb38402e2a321d85]
  module.myapp-webserver[0].aws_instance.myapp-server: Refreshing state... [id=i-0f5c904ff54acbb52]

  Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following
  symbols:
    ~ update in-place
    - destroy

  Terraform will perform the following actions:

    # module.myapp-webserver[0].aws_security_group.web_sg will be updated in-place
    ~ resource "aws_security_group" "web_sg" {
          id                     = "sg-088adec1554a602fa"
        ~ ingress                = [
            - {
                - cidr_blocks      = [
                    - "0.0.0.0/0",
                  ]
                - from_port        = 22
                - ipv6_cidr_blocks = []
                - prefix_list_ids  = []
```

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform apply -auto-approve
    ]
module.myapp-webserver[1].aws_instance.myapp-server: Destroying... [id=i-09fbfd54e203f408d]
module.myapp-webserver[2].aws_instance.myapp-server: Destroying... [id=i-0fb38402e2a321d85]
module.myapp-webserver[0].aws_security_group.web_sg: Modifying... [id=sg-088adec1554a602fa]
module.myapp-webserver[0].aws_security_group.web_sg: Modifications complete after 1s [id=sg-088adec1554a602fa]
module.myapp-webserver[1].aws_instance.myapp-server: Still destroying... [id=i-09fbfd54e203f408d, 00m10s elapsed]
module.myapp-webserver[2].aws_instance.myapp-server: Still destroying... [id=i-0fb38402e2a321d85, 00m10s elapsed]
module.myapp-webserver[1].aws_instance.myapp-server: Still destroying... [id=i-09fbfd54e203f408d, 00m20s elapsed]
module.myapp-webserver[2].aws_instance.myapp-server: Still destroying... [id=i-0fb38402e2a321d85, 00m20s elapsed]
module.myapp-webserver[1].aws_instance.myapp-server: Destruction complete after 30s
module.myapp-webserver[1].aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-1]
module.myapp-webserver[1].aws_security_group.web_sg: Destroying... [id=sg-039ed491d98cbaaff]
module.myapp-webserver[2].aws_instance.myapp-server: Still destroying... [id=i-0fb38402e2a321d85, 00m30s elapsed]
module.myapp-webserver[1].aws_key_pair.ssh-key: Destruction complete after 0s
module.myapp-webserver[1].aws_security_group.web_sg: Destruction complete after 1s
module.myapp-webserver[2].aws_instance.myapp-server: Still destroying... [id=i-0fb38402e2a321d85, 00m40s elapsed]
module.myapp-webserver[2].aws_instance.myapp-server: Still destroying... [id=i-0fb38402e2a321d85, 00m50s elapsed]
module.myapp-webserver[2].aws_instance.myapp-server: Still destroying... [id=i-0fb38402e2a321d85, 01m00s elapsed]
module.myapp-webserver[2].aws_instance.myapp-server: Still destroying... [id=i-0fb38402e2a321d85, 01m10s elapsed]
module.myapp-webserver[2].aws_instance.myapp-server: Still destroying... [id=i-0fb38402e2a321d85, 01m20s elapsed]
module.myapp-webserver[2].aws_instance.myapp-server: Destruction complete after 1m21s
module.myapp-webserver[2].aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-2]
module.myapp-webserver[2].aws_security_group.web_sg: Destroying... [id=sg-0a018c86feda1c480]
module.myapp-webserver[2].aws_key_pair.ssh-key: Destruction complete after 0s
module.myapp-webserver[2].aws_security_group.web_sg: Destruction complete after 1s

Apply complete! Resources: 0 added, 1 changed, 6 destroyed.

Outputs:

webserver_public_ips = [
  "3.29.33.227",
]
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

4. Outputs:

task5_terraform_output_single_ip.png — single EC2 public IP.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform output
webserver_public_ips = [
  "3.29.33.227",
]
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

5. Adjust hosts:

task5_hosts_nginx_group.png — hosts file for [nginx].

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

[nginx]
3.29.33.227

[nginx:vars]
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_user=ec2-user

~
~
~
~
```

6. Update my-playbook.yaml:

task5_my_playbook_nginx_group.png — playbook with hosts: nginx.

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```yaml
---
- name: Configure nginx web server
  hosts: nginx
  become: true
  tasks:
    - name: install nginx and update cache
      yum:
        name: nginx
        state: present
        update_cache: yes

    - name: install openssl
      yum:
        name: openssl
        state: present

    - name: start nginx server
      service:
        name: nginx
        state: started
        enabled: true

~
~
```

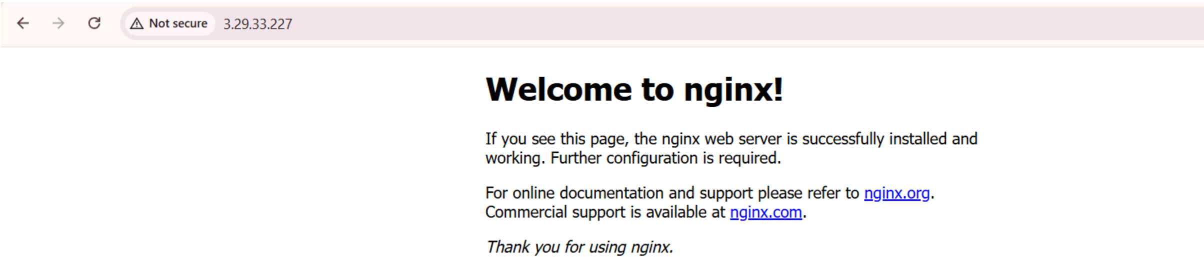
7. Run the playbook:

task5_ansible_play_nginx_group.png — playbook output.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml
[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an ansible.cfg source. F
 more information see https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writable-dir

PLAY [Configure nginx web server] ***************************************************************************************************

TASK [Gathering Facts] **************************************************************************************************************
[WARNING]: Host '3.29.33.227' is using the discovered Python interpreter at '/usr/bin/python3.9', but future installation of another P
hon interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendi
s/interpreter_discovery.html for more information.
ok: [3.29.33.227]

TASK [install nginx and update cache] ***********************************************************************************************
ok: [3.29.33.227]

TASK [install openssl] **************************************************************************************************************
ok: [3.29.33.227]

TASK [start nginx server] ***********************************************************************************************************
changed: [3.29.33.227]

PLAY RECAP **************************************************************************************************************************
3.29.33.227                : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

8. **Verify nginx default page** at http://<public-ip>

task5_nginx_browser_single.png — browser view.

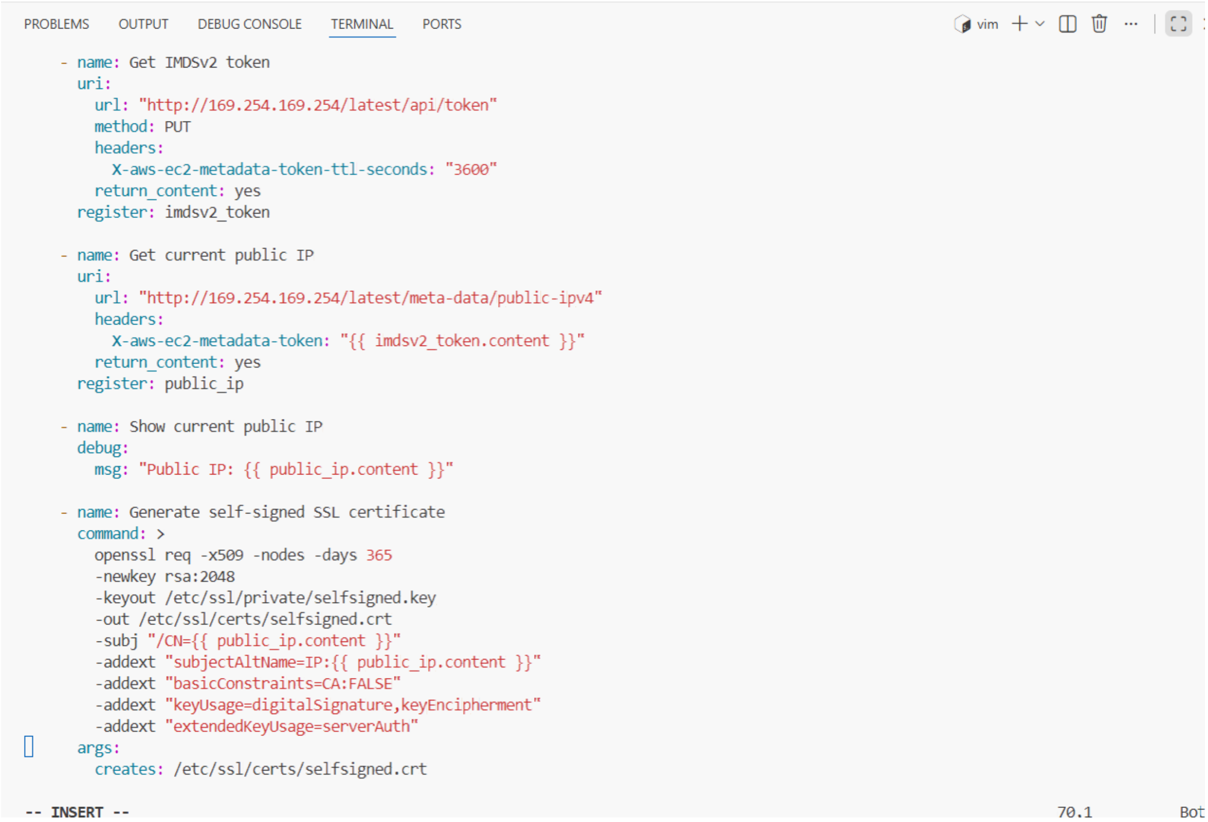


Task 6 - Ansible-managed SSL certificates

Extend your playbook to generate **self-signed SSL certificates** using dynamic public IP.

1. Append this play after the nginx play in my-playbook.yaml:

task6_my_playbook_ssl_section.png — SSL block in my-playbook.yaml.




```
  - name: Get IMDSv2 token
    uri:
      url: "http://169.254.169.254/latest/api/token"
      method: PUT
      headers:
        X-aws-ec2-metadata-token-ttl-seconds: "3600"
      return_content: yes
    register: imdsv2_token

  - name: Get current public IP
    uri:
      url: "http://169.254.169.254/latest/meta-data/public-ipv4"
      headers:
        X-aws-ec2-metadata-token: "{{ imdsv2_token.content }}"
      return_content: yes
    register: public_ip

  - name: Show current public IP
    debug:
      msg: "Public IP: {{ public_ip.content }}"

  - name: Generate self-signed SSL certificate
    command: >
      openssl req -x509 -nodes -days 365
      -newkey rsa:2048
      -keyout /etc/ssl/private/selfsigned.key
      -out /etc/ssl/certs/selfsigned.crt
      -subj "/CN={{ public_ip.content }}"
      -addext "subjectAltName=IP:{{ public_ip.content }}"
      -addext "basicConstraints=CA:FALSE"
      -addext "keyUsage=digitalSignature,keyEncipherment"
      -addext "extendedKeyUsage=serverAuth"
    args:
      creates: /etc/ssl/certs/selfsigned.crt
-- INSERT --                                                70.1          Bot
```

2. Run:

task6_ansible_play_ssl.png — SSL tasks output.

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                        bash  +  ⌄  ⊞  🗑  ···  |  ⛶  ×
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml
TASK [install openssl] ***************************************************************************************************
ok: [3.29.33.227]

TASK [start nginx server] ************************************************************************************************
ok: [3.29.33.227]

PLAY [Configure SSL certificates] ****************************************************************************************

TASK [Gathering Facts] ***************************************************************************************************
ok: [3.29.33.227]

TASK [Create SSL private directory] **************************************************************************************
changed: [3.29.33.227]

TASK [Create SSL certs directory] ****************************************************************************************
changed: [3.29.33.227]

TASK [Get IMDSv2 token] **************************************************************************************************
ok: [3.29.33.227]

TASK [Get current public IP] *********************************************************************************************
ok: [3.29.33.227]

TASK [Show current public IP] ********************************************************************************************
ok: [3.29.33.227] => {
    "msg": "Public IP: 3.29.33.227"
}

TASK [Generate self-signed SSL certificate] ******************************************************************************
changed: [3.29.33.227]

PLAY RECAP ***************************************************************************************************************
3.29.33.227                : ok=11   changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

○ @23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ 
```

3. SSH and verify:

task6_ssl_cert_file.png — cat /etc/ssl/certs/selfsigned.crt.

```
○ @23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ ssh -i ~/.ssh/id_ed25519 ec2-user@3.29.33.227
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sat Jan 10 13:24:01 2026 from 4.240.18.225
[ec2-user@ip-10-0-10-112 ~]$ sudo cat /etc/ssl/certs/selfsigned.crt
-----BEGIN CERTIFICATE-----
```

```
MIIDOzCCAiOgAwIBAgIUQ928d+6Hghyr6JfCX4rM6FpcU7UwDQYJKoZIhvcNAQEL
BQAwFjEUMBIGA1UEAwwLMy4yOS4zMy4yMjcwHhcNMjYwMTEwMTMyNDAxWhcNMjcw
MTEwMTMyNDAxWjAWMRQwEgYDVQQDDAszLjI5LjMzLjIyNzCCASIwDQYJKoZIhvcN
AQEBBQADggEPADCCAQoCggEBAMyFHNvNHdo3ecnlThLQiPZihnybwyiS9IDocv9V
dQkubc3YtKFbWgvEiDDiOPhozwMk9AX7rNBvobuuPGIwAZdf7knZAac+fZe7Ytcn
Xf0elxff5Npj46gywzCGDQGLdMcUMJCVn62lJv8FC6bnplqfqYR803zY0oOezCv3
wd/HotC4tjeWv/B39oBLITniwy9RbafH3pLbCDhzU3wl3AV3cWZoQ5pKjIxHQLPI
6i9yuA4+gpChu4bHkDyQL8j6DO2kHLalYf2S3DPCPC/jiBE1N7kVmS2HCyi2NdZW
ryZp3Rzn7OHK/lxG/3GOmak0Tyd+8mDWa0PRE2p56QGvfBMCAwEAAaOBgDB+MB0G
A1UdDgQWBBS7OUrpXFy5H/A0Flgu4LWahzlqxzAfBgNVHSMEGDAWgBS7OUrpXFy5
H/A0Flgu4LWahzlqxzAPBgNVHREECDAGhwQDHSHjMAkGA1UdEwQCMAAwCwYDVR0P
BAQDAgWgMBMGA1UdJQQMMAoGCCsGAQUFBwMBMA0GCSqGSIb3DQEBCwUAA4IBAQBT
QI/85cjoewoXLNIByVvyRQkhq+cG1kZ3L8bq0KE7YfjSbLc2QzxwUowUTS57A1RE
Hassume7L/lIHJtT903D/l7B6gZoHM0lpgwz7eLNJFC8WtGJyx5VFeZqos7zp0W2
xdM4to4QG/qF+ABDcfZYA2CPWpZbTvWzu+lCGoPxwISj+rNFEzsk5H7yCYo2KXDM
E+DzaKlkiwiinoLE5f00PVwjLYb8xmEfohSjNxTsiyrPb+Gk0fuZxSLxfgplfRG0
13TEvrzc9bHZOPQ/esG8OANPJnT1h2zXZi5oTuXprixqIH50qpp75vPQfUvqXZva
mj+nrIZyxUSY51TJ0x+k
```

```
-----END CERTIFICATE-----
[ec2-user@ip-10-0-10-112 ~]$ 
```

task6_ssl_key_file.png — cat /etc/ssl/private/selfsigned.key.

```
[ec2-user@ip-10-0-10-112 ~]$ sudo cat /etc/ssl/private/selfsigned.key
```

```
-----BEGIN PRIVATE KEY-----
MIIEvgIBADANBgkqhkiG9w0BAQEFAASCBKgwggSkAgEAAoIBAQDMhRzbzR3aN3nJ
5U4S0Ij2YoZ8m8MokvSA6HL/VXUJLm3N2LShW1oLxIgw4jj4aM8DJPQF+6zQb6G7
rjxiMAGXX+5J2QGnPn2Xu2LXJ139HpcX3+TaY+OoMsMwhg0Bi3THFDCQlZ+tpSb/
BQum56Zan6mEfNN82NKDnswr98Hfx6LQuLY3lr/wd/aASyE54sMvUW2nx96S2wg4
c1N8JdwFd3FmaEOaSoyMR0CzyOovcrgOPoKQobuGx5A8kC/I+gztpBy2pWH9ktwz
wjwv44gRNTe5FZkthwsotjXWVq8mad0c5+zhyv5cRv9xjpmpNE8nfvJg1mtD0RNq
eekBr3wTAgMBAAECggEAGjr4TeMrcPwBzWMbdaOel/sK70W1l1PsXUbII6H84LCT
uf2RLSC0681zqiau0oKetPMzA5FubffjRlmSZQVt2XxJk4RsMdxR6flJM/mt6Z4t
bWJ0d44Cz2UIo7CNpSUS/8rBxR7A3AR9Jo1UcT5El7A34vDKLw5KnEGW7gyt+4U5
dtSW/zAAkIRKZ7oB6wE9GunhEnN8dd7dIMCJORvCP5SOC6QCSO+JvaEem6wQYrx/
iP3Mko/6LGKtxS7HSTMu4wKH6kAXQALcAfMnMbnRLIcx16j2pO2JLsBD5ETnGw37
vuzfctlduumxirJJ0iMfE2JAoS8du/wQ5YEXq4a0AQKBgQD//w5dzME8GyXBMZvm
+t2m2jjGMyqtdFyBSHTJegbASIEMJUBv1dk6e6QJRk5IwNswsdbRVCjtBW/Iji8e
g5bSLr6rGi25bAwseyI8fxQ6CK3tDRAXKjFFhmEmbH300x/iK7anwBAJHULlWHOb
hT7p7YrFhvjTaYSrsKBKp5l+BQKBgQDMhd3naKvwUzlZAJSeerNSDbyDLncNbEgE
//v2D0oiDXx6XYT3bxuijq6zoHvjQV31ADjnUQtC66wZonbHcb0ircwYiBTZ8jFd
tOodwVj2cv6R2qL15P6Z46s7yHESXgRpp9VhZ3jUJCZ76eHXBVnoHVMUDlcg75qh
1JduNhMVNwKBgQDfn01McQ2n1MshPZXDqwcuttf7gzzXspsQMFwcl+UH2Q4pruzg
BL2y0cjUdT7zXMzcd62AxHTPv/0LA/FzRV3dkwnp639fx1vHC3U3xYbEv0o3lj6l
Da1m32FEMGfDEx558tenKCaZ357NO3wZscbokCG2NRZtj3JejHcpX8i8hQKBgQCB
PqtfXFcjdQRNiT9h8vo1BTc20vsrAV11w2qvMTRQ1qLuFQE89/rHvFEMdIM1to4w
kGZlT2LFdHb6cB+NpAO40YmjiwNaQ2IM7JHyIcgdXcgf1cGbkI2QJD6GdgF2cPFq
70O53RCPRxUGw2MTZW9ObpGcn/9RcfkPAXB8ljFy+wKBgCzfFRRIfoGzshW2fiGG
wkao3oXmCBiBZVB8sEAfI3ugPxBRy+j2hDwgFyc8FAwx0DnqhSlEUvuQqg4SrPQu
pC4VvUWVxBfA6L8I+Mlmy7f3iVL2QOAIXg3A3w6NiG2kh169av+1DOlVFQpa4Jqj
1+whdPn1jUuxhfbacGkFLWE5
-----END PRIVATE KEY-----
```

```
[ec2-user@ip-10-0-10-112 ~]$
```

Task 7 - PHP front-end deployment with templates

Deploy a PHP application and Nginx configuration using Ansible copy and template.

1. **Create directories and files**:

task7_files_templates_created.png — ls -R showing files/index.php and templates/nginx.conf.j2

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                   bash
_elementtree.cpython-313-x86_64-linux-gnu.so      grp.cpython-313-x86_64-linux-gnu.so
_hashlib.cpython-313-x86_64-linux-gnu.so          math.cpython-313-x86_64-linux-gnu.so
_heapq.cpython-313-x86_64-linux-gnu.so            mmap.cpython-313-x86_64-linux-gnu.so
_json.cpython-313-x86_64-linux-gnu.so             pyexpat.cpython-313-x86_64-linux-gnu.so
_lzma.cpython-313-x86_64-linux-gnu.so             readline.cpython-313-x86_64-linux-gnu.so
_md5.cpython-313-x86_64-linux-gnu.so              resource.cpython-313-x86_64-linux-gnu.so
_multibytecodec.cpython-313-x86_64-linux-gnu.so   select.cpython-313-x86_64-linux-gnu.so
_multiprocessing.cpython-313-x86_64-linux-gnu.so  termios.cpython-313-x86_64-linux-gnu.so
_opcode.cpython-313-x86_64-linux-gnu.so           unicodedata.cpython-313-x86_64-linux-gnu.so
_pickle.cpython-313-x86_64-linux-gnu.so           zlib.cpython-313-x86_64-linux-gnu.so
_posixshmem.cpython-313-x86_64-linux-gnu.so

./aws/dist/prompt_toolkit-3.0.51.dist-info:
INSTALLER  METADATA  RECORD  WHEEL  licenses  top_level.txt

./aws/dist/prompt_toolkit-3.0.51.dist-info/licenses:
AUTHORS.rst  LICENSE

./aws/dist/wheel-0.45.1.dist-info:
INSTALLER  LICENSE.txt  METADATA  RECORD  REQUESTED  WHEEL  direct_url.json  entry_points.txt

./files:
index.php

./modules:
subnet  webserver

./modules/subnet:
main.tf  outputs.tf  variables.tf

./modules/webserver:
main.tf  outputs.tf  variables.tf

./templates:
nginx.conf.j2
@23-22411-057-sudo ➜ /workspaces/terraform machine (main) $
```

.

2. Fill files/index.php with the following PHP metadata page:

task7_index_php_content.png — content of files/index.php in editor or cat.

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                   nano
  GNU nano 7.2                                files/index.php *
<?php
// Get hostname
$hostname = gethostname();

// Deployment date
$deployed_date = date("Y-m-d H:i:s");

// Metadata base URL
$metadata_base = "http://169.254.169.254/latest/";

// Function to get IMDSv2 token
function getImdsV2Token() {
    $ch = curl_init("http://169.254.169.254/latest/api/token");
    curl_setopt_array($ch, [
        CURLOPT_RETURNTRANSFER => true,
        CURLOPT_CUSTOMREQUEST  => "PUT",
        CURLOPT_HTTPHEADER     => [
            "X-aws-ec2-metadata-token-ttl-seconds: 21600"
        ],
        CURLOPT_TIMEOUT        => 2
    ]);

    $token = curl_exec($ch);
    curl_close($ch);

    return $token ?: null;
}

// Function to fetch metadata using token
function getMetadata($path, $token) {
    $url = "http://169.254.169.254/latest/meta-data/" . $path;
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS nano

```
 GNU nano 7.2                                   files/index.php *
    $ch = curl_init($url);
    curl_setopt_array($ch, [
        CURLOPT_RETURNTRANSFER => true,
        CURLOPT_HTTPHEADER     => [
            "X-aws-ec2-metadata-token: $token"
        ],
        CURLOPT_TIMEOUT        => 2
    ]);

    $value = curl_exec($ch);
    curl_close($ch);

    return $value ?: "N/A";
}

// Fetch token
$token = getImdsV2Token();

// Fetch metadata only if token is available
$instance_id = $token ? getMetadata("instance-id", $token) : "N/A";
$private_ip  = $token ? getMetadata("local-ipv4", $token) : "N/A";
$public_ip   = $token ? getMetadata("public-ipv4", $token) : "N/A";
$public_dns  = $token ? getMetadata("public-hostname", $token) : "N/A";
?>
<!DOCTYPE html>
<html>
<head>
    <title>Frontend Web Server</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 50px;
```

3. Fill templates/nginx.conf.j2 with the following Nginx configuration template:

task7_nginx_conf_template.png — content of templates/nginx.conf.j2.

```
 GNU nano 7.2                                   templates/nginx.conf.j2 *
user nginx;
worker_processes auto;
error_log /var/log/nginx/error.log notice;
pid /run/nginx.pid;

events {
    worker_connections 1024;
}

http {
    log_format  main  '$remote_addr - $remote_user [$time_local] "$request"'
                      '$status $body_bytes_sent "$http_referer"'
                      '"$http_user_agent" "$http_x_forwarded_for"';

    access_log  /var/log/nginx/access.log  main;

    sendfile            on;
    tcp_nopush          on;
    keepalive_timeout   65;
    types_hash_max_size 4096;

    include             /etc/nginx/mime.types;
    default_type        application/octet-stream;

    upstream backend_servers {
        server 158.252.94.241:80;
        server 158.252.94.242:80 backup;
    }

    server {
        listen 443 ssl;
        server_name {{ server_public_ip }};
```

```
GNU nano 7.2                         templates/nginx.conf.j2 *
        server 158.252.94.242:80 backup;
    }

    server {
        listen 443 ssl;
        server_name {{ server_public_ip }};
        ssl_certificate /etc/ssl/certs/selfsigned.crt;
        ssl_certificate_key /etc/ssl/private/selfsigned.key;

        location / {
            root /usr/share/nginx/html;
            index index.php index.html index.htm;
    #       proxy_pass http://158.252.94.241:80;
    #       proxy_pass http://backend_servers;

            # 🔴 This block is necessary for Php Website
            location ~ \.php$ {
                include fastcgi_params;
                fastcgi_pass unix:/run/php-fpm/www.sock;
                fastcgi_index index.php;
                fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
            }
        }
    }

    server {
        listen 80;
        server_name _;
        return 301 https://$host$request_uri;
    }
}
```

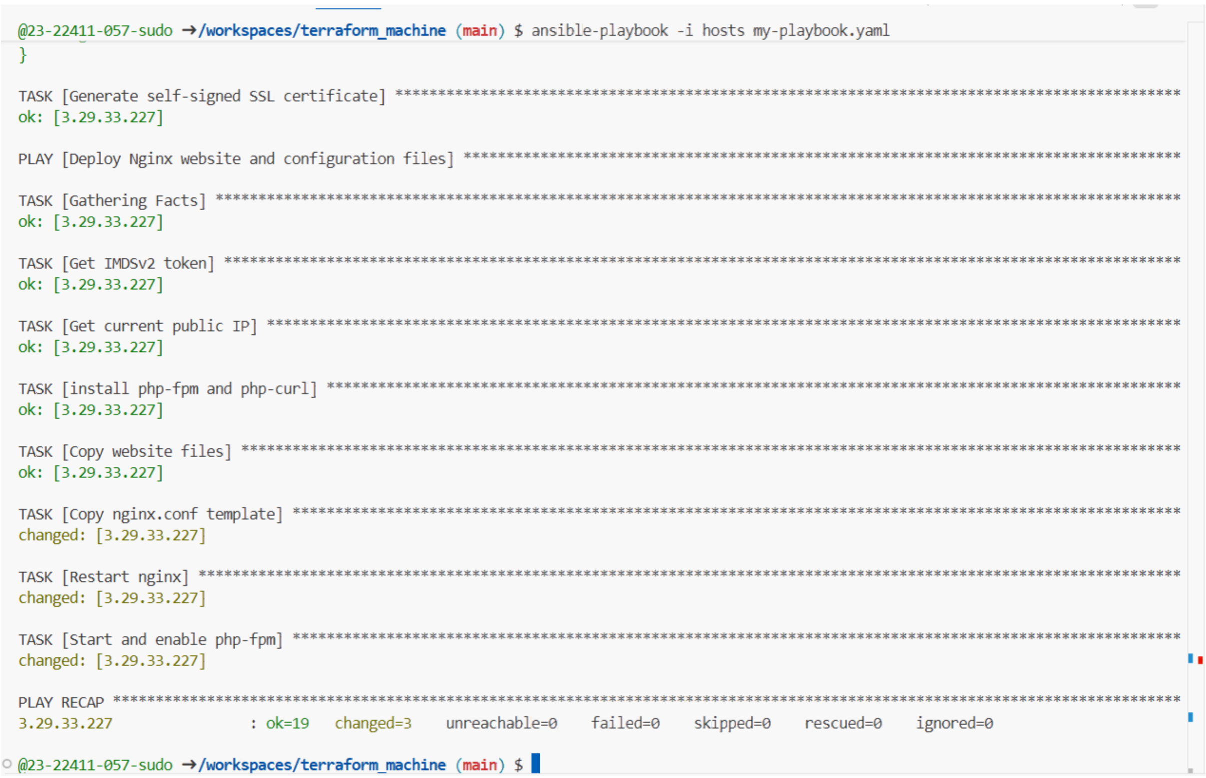
4. **Add this play to my-playbook.yaml**:

task7_my_playbook_web_deploy.png — updated my-playbook.yaml including this play.

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                    nano
GNU nano 7.2                         my-playbook.yaml *
- name: Deploy Nginx website and configuration files
  hosts: nginx
  become: true
  tasks:
    - name: install php-fpm and php-curl
      yum:
        name:
          - php-fpm
          - php-curl
        state: present

    - name: Copy website files
      copy:
        src: files/index.php
        dest: /usr/share/nginx/html/index.php
        owner: nginx
        group: nginx
        mode: '0644'

    - name: Copy nginx.conf template
      template:
        src: templates/nginx.conf.j2
        dest: /etc/nginx/nginx.conf
        owner: root
        group: root
        mode: '0644'

    - name: Restart nginx
      service:
        name: nginx
        state: restarted
```

5. **Run the playbook**:

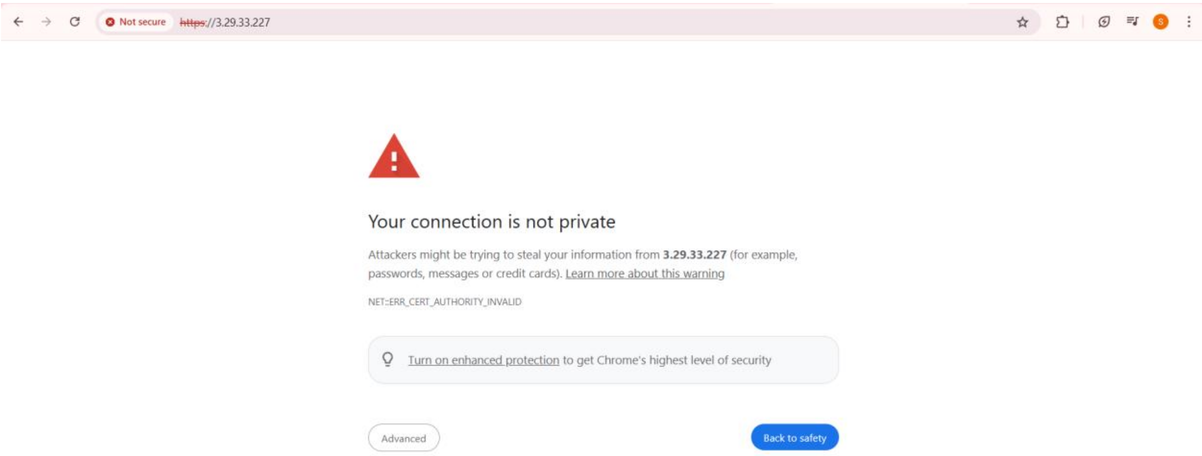
task7_ansible_play_web_deploy.png — playbook output showing PHP/nginx tasks executed.

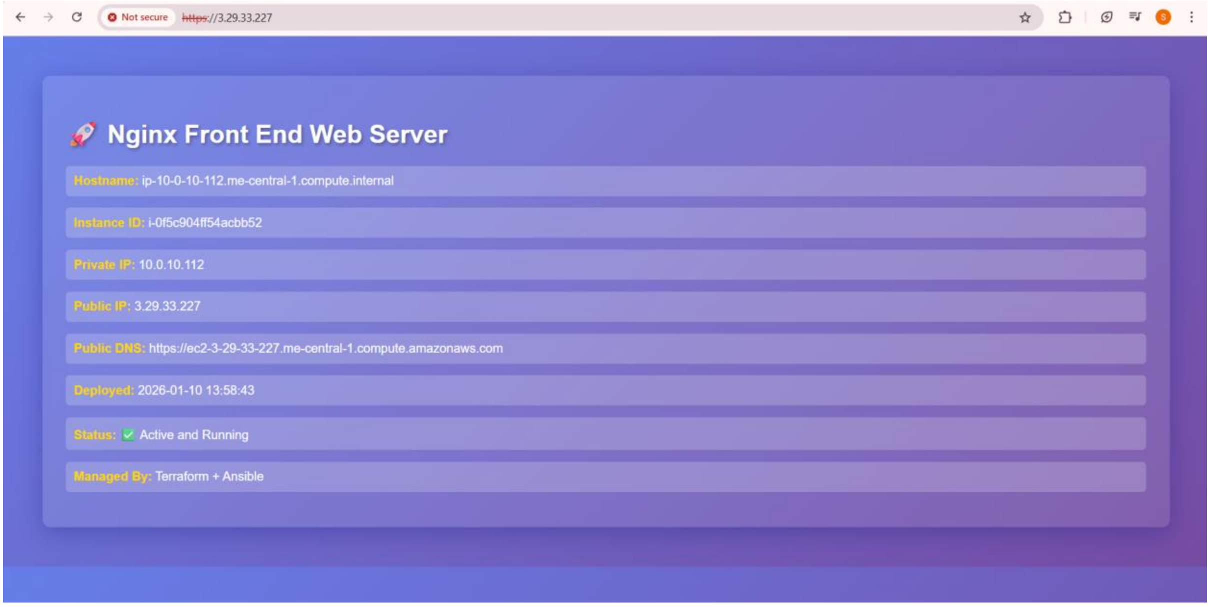


```
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml
}

TASK [Generate self-signed SSL certificate] ****************************************************************************************
ok: [3.29.33.227]

PLAY [Deploy Nginx website and configuration files] ********************************************************************************

TASK [Gathering Facts] *************************************************************************************************************
ok: [3.29.33.227]

TASK [Get IMDSv2 token] ************************************************************************************************************
ok: [3.29.33.227]

TASK [Get current public IP] *******************************************************************************************************
ok: [3.29.33.227]

TASK [install php-fpm and php-curl] ************************************************************************************************
ok: [3.29.33.227]

TASK [Copy website files] **********************************************************************************************************
ok: [3.29.33.227]

TASK [Copy nginx.conf template] ****************************************************************************************************
changed: [3.29.33.227]

TASK [Restart nginx] ***************************************************************************************************************
changed: [3.29.33.227]

TASK [Start and enable php-fpm] ****************************************************************************************************
changed: [3.29.33.227]

PLAY RECAP *************************************************************************************************************************
3.29.33.227                : ok=19   changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $
```

6. Visit https://<public-ip>:

task7_php_https_browser.png — browser showing PHP web page via HTTPS.





Task 8 – Docker & Docker Compose provisioning via Ansible

Deploy Docker and Docker Compose on a new EC2 instance.

1. **Destroy old infrastructure**:

task8_terraform_destroy_old.png — destroy output for old stack.

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                bash
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform destroy -auto-approve

Plan: 0 to add, 0 to change, 7 to destroy.

Changes to Outputs:
  - webserver_public_ips = [
      - "3.29.33.227",
    ] -> null
module.myapp-subnet.aws_default_route_table.main_rt: Destroying... [id=rtb-0cd5a79d0f7e93439]
module.myapp-webserver[0].aws_instance.myapp-server: Destroying... [id=i-0f5c904ff54acbb52]
module.myapp-subnet.aws_default_route_table.main_rt: Destruction complete after 0s
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destroying... [id=igw-01507100278956522]
module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0f5c904ff54acbb52, 00m10s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-01507100278956522, 00m10s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0f5c904ff54acbb52, 00m20s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-01507100278956522, 00m20s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0f5c904ff54acbb52, 00m30s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-01507100278956522, 00m30s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0f5c904ff54acbb52, 00m40s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-01507100278956522, 00m40s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0f5c904ff54acbb52, 00m50s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-01507100278956522, 00m50s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destruction complete after 58s
module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0f5c904ff54acbb52, 01m00s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Destruction complete after 1m1s
module.myapp-webserver[0].aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-0]
module.myapp-webserver[0].aws_security_group.web_sg: Destroying... [id=sg-088adec1554a602fa]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destroying... [id=subnet-09842c89977df0aa4]
module.myapp-webserver[0].aws_key_pair.ssh-key: Destruction complete after 0s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 1s
module.myapp-webserver[0].aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0bf2e8de0352486a5]
aws_vpc.myapp_vpc: Destruction complete after 0s

Destroy complete! Resources: 7 destroyed.
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

2. **Recreate fresh infrastructure** (1 instance):

task8_terraform_apply_docker_instance.png — new apply.

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                nano

 GNU nano 7.2                                main.tf
  tags = {
     Name = "${var.env_prefix}-vpc"
  }
}

module "myapp-subnet" {
  source = "./modules/subnet"
  vpc_id = aws_vpc.myapp_vpc.id
  subnet_cidr_block = var.subnet_cidr_block
  availability_zone = var.availability_zone
  env_prefix = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}

module "myapp-webserver" {
  source            = "./modules/webserver"
  env_prefix        = var.env_prefix
  instance_type     = var.instance_type
  availability_zone = var.availability_zone
  public_key        = var.public_key
  my_ip             = local.my_ip
  vpc_id            = aws_vpc.myapp_vpc.id
  subnet_id         = module.myapp-subnet.subnet.id

  # Loop count
  count = 1

  # Use count.index to differentiate instances
  instance_suffix = count.index
}
```

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform apply -auto-approve
      + vpc_id                 = (known after apply)
    }

Plan: 7 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + webserver_public_ips = [
      + (known after apply),
    ]
module.myapp-webserver[0].aws_key_pair.ssh-key: Creating...
aws_vpc.myapp_vpc: Creating...
module.myapp-webserver[0].aws_key_pair.ssh-key: Creation complete after 0s [id=dev-serverkey-0]
aws_vpc.myapp_vpc: Still creating... [00m10s elapsed]
aws_vpc.myapp_vpc: Creation complete after 12s [id=vpc-0ade7c3c4a67a6421]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Creating...
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creating...
module.myapp-webserver[0].aws_security_group.web_sg: Creating...
module.myapp-subnet.aws_internet_gateway.myapp_igw: Creation complete after 0s [id=igw-0cf955965564dac4c]
module.myapp-subnet.aws_default_route_table.main_rt: Creating...
module.myapp-subnet.aws_default_route_table.main_rt: Creation complete after 1s [id=rtb-0b063f601309188d0]
module.myapp-webserver[0].aws_security_group.web_sg: Creation complete after 3s [id=sg-05d304624bf559a84]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Still creating... [00m10s elapsed]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creation complete after 11s [id=subnet-0ccedd3e703d0f611]
module.myapp-webserver[0].aws_instance.myapp-server: Creating...
module.myapp-webserver[0].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Creation complete after 12s [id=i-0782572662a34a57c]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:

webserver_public_ips = [
  "158.252.83.114",
]
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

task8_terraform_output_new_ip.png — output showing new instance IP.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform output
webserver_public_ips = [
  "158.252.83.114",
]
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

3. **Update hosts**:

task8_hosts_docker_servers.png — hosts file showing [docker_servers].

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS vim

```
[docker_servers]
158.252.83.114

[docker_servers:vars]
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_user=ec2-user

~
~
~
```

4. **Replace my-playbook.yaml** content with:

task8_my_playbook_docker.png — new Docker playbook content.

```
GNU nano 7.2                    my-playbook.yaml *
---
- name: Configure Docker
  hosts: all
  become: true
  tasks:
    - name: install docker and update cache
      yum:
        name: docker
        state: present
        update_cache: yes

- name: Install Docker Compose
  hosts: all
  become: true
  gather_facts: true
  tasks:
    - name: create docker cli-plugins directory
      file:
        path: /usr/local/lib/docker/cli-plugins
        state: directory
        mode: '0755'

    - name: install docker-compose
      get_url:
        url: https://github.com/docker/compose/releases/latest/download/docker-compose-linux-{{ lookup('pipe', 'uname -m') }}
        dest: /usr/local/lib/docker/cli-plugins/docker-compose
        mode: +x

    - name: View architecture of the system
      debug:
        msg: "System architecture of {{ inventory_hostname }} is {{ ansible_facts['architecture'] }}"

^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

5. **Run the play**:

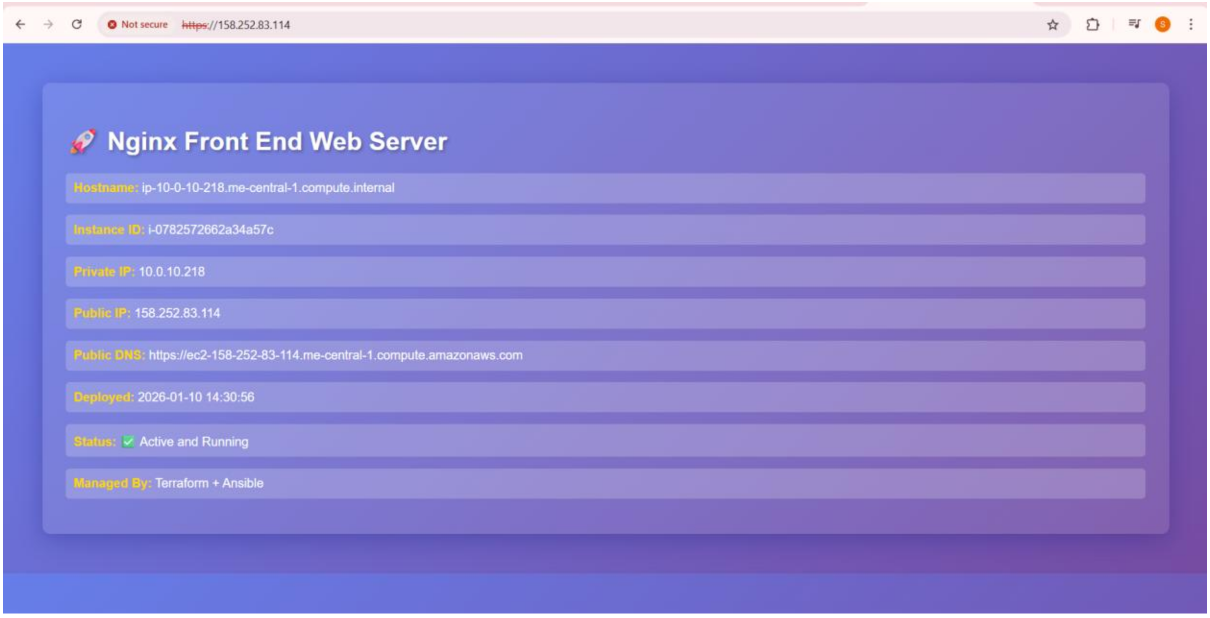
task8_ansible_play_docker.png — playbook output installing Docker and Compose.

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                   bash
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml

TASK [Gathering Facts] *******************************************************************************************
ok: [158.252.83.114]

TASK [install docker and update cache] ***************************************************************************
changed: [158.252.83.114]

PLAY [Install Docker Compose] ************************************************************************************

TASK [Gathering Facts] *******************************************************************************************
ok: [158.252.83.114]

TASK [create docker cli-plugins directory] ***********************************************************************
changed: [158.252.83.114]

TASK [install docker-compose] ************************************************************************************
changed: [158.252.83.114]

TASK [View architecture of the system] ***************************************************************************
ok: [158.252.83.114] => {
    "msg": "System architecture of 158.252.83.114 is x86_64"
}

TASK [Alternate method to view architecture of the system] *******************************************************
ok: [158.252.83.114] => {
    "msg": "System architecture of 158.252.83.114 is x86_64"
}

TASK [restart docker service] ************************************************************************************
changed: [158.252.83.114]

PLAY RECAP *******************************************************************************************************
158.252.83.114             : ok=25   changed=7    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $
```

6. **Verify Docker**:

task8_docker_ps_remote.png — sudo docker ps output from EC2 instance.

```
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ ssh -i ~/.ssh/id_ed25519 ec2-user@158.252.83.114
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Sat Jan 10 14:30:24 2026 from 4.240.18.225
[ec2-user@ip-10-0-10-218 ~]$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[ec2-user@ip-10-0-10-218 ~]$ docker --version
Docker version 25.0.13, build 0bab007
```

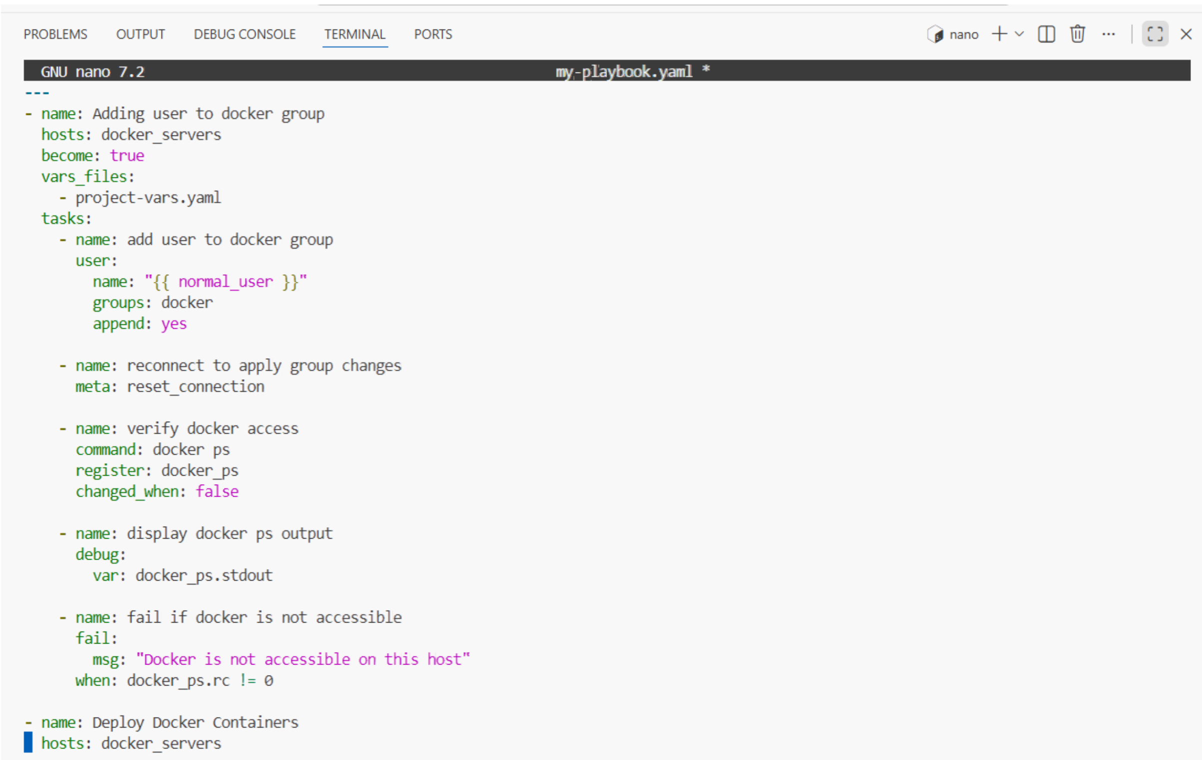


Task 9 – Gitea Docker stack via Ansible + Terraform security group update

Run containers for **Gitea + Postgres** and open port 3000 in the security group.

1. **Extend my-playbook.yaml** by appending:

task9_my_playbook_add_user_to_docker.png — new play appended in my-playbook.yaml.



```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                nano
GNU nano 7.2                         my-playbook.yaml *
---
- name: Adding user to docker group
  hosts: docker_servers
  become: true
  vars_files:
    - project-vars.yaml
  tasks:
    - name: add user to docker group
      user:
        name: "{{ normal_user }}"
        groups: docker
        append: yes

    - name: reconnect to apply group changes
      meta: reset_connection

    - name: verify docker access
      command: docker ps
      register: docker_ps
      changed_when: false

    - name: display docker ps output
      debug:
        var: docker_ps.stdout

    - name: fail if docker is not accessible
      fail:
        msg: "Docker is not accessible on this host"
      when: docker_ps.rc != 0

- name: Deploy Docker Containers
  hosts: docker_servers
```

2. **Create project-vars.yaml**:

task9_project_vars.png — content of project-vars.yaml.

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                  nano
GNU nano 7.2                                   project-vars.yaml *
normal_user: ec2-user
docker_compose_file_location: /workspaces/terraform_machine   # Path where compose.yaml exists
```

3. Add the deploy containers play:

task9_my_playbook_deploy_containers.png — final my-playbook.yaml with Docker deploy play.

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                  nano
GNU nano 7.2                                   my-playbook.yaml *
      debug:
        var: docker_ps.stdout

    - name: fail if docker is not accessible
      fail:
        msg: "Docker is not accessible on this host"
      when: docker_ps.rc != 0

- name: Deploy Docker Containers
  hosts: docker_servers
  become: true
  user: "{{ normal_user }}"
  vars_files:
    - project-vars.yaml
  tasks:
    - name: check if docker-compose file exists
      stat:
        path: /home/{{ normal_user }}/compose.yaml
      register: compose_file

    - name: copy docker-compose file
      copy:
        src: "{{ docker_compose_file_location }}/compose.yaml"
        dest: /home/{{ normal_user }}/compose.yaml
        mode: '0644'
      when: not compose_file.stat.exists

    - name: deploy containers using docker-compose
      command: docker compose -f /home/{{ normal_user }}/compose.yaml up -d
      register: compose_result
      changed_when: "'Creating' in compose_result.stdout or 'Recreating' in compose_result.stdout"
```

4. Create compose.yaml in repo root:

task9_compose_yaml.png — content of compose.yaml.

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                nano  +  ⌵  ◫  🗑  ···  ⛶  ×

 GNU nano 7.2                                    compose.yaml *
version: '3.8'

services:
  gitea:
    image: gitea/gitea:latest
    container_name: gitea
    environment:
      - DB_TYPE=postgres
      - DB_HOST=db:5432
      - DB_NAME=gitea
      - DB_USER=gitea
      - DB_PASSWD=gitea
    restart: always
    volumes:
      - gitea:/data
    ports:
      - 3000:3000
    networks:
      - webnet

  db:
    image: postgres:alpine
    container_name: gitea_db
    environment:
      - POSTGRES_USER=gitea
      - POSTGRES_PASSWORD=gitea
      - POSTGRES_DB=gitea
    restart: always
    volumes:
      - gitea_postgres:/var/lib/postgresql/data
    expose:
      - 5432
```

5. **Run playbook and visit the public-ip on your browser but webpage will not be shown due to permission issue:**

task9_ansible_play_gitea.png — playbook output showing Gitea/db containers created.

```
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml

TASK [add user to docker group] ***************************************************************************
changed: [158.252.83.114]

TASK [reconnect to apply group changes] *******************************************************************

TASK [verify docker access] *******************************************************************************
ok: [158.252.83.114]

TASK [display docker ps output] ***************************************************************************
ok: [158.252.83.114] => {
    "docker_ps.stdout": "CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES"
}

TASK [fail if docker is not accessible] *******************************************************************
skipping: [158.252.83.114]

PLAY [Deploy Docker Containers] ***************************************************************************

TASK [Gathering Facts] ************************************************************************************
ok: [158.252.83.114]

TASK [check if docker-compose file exists] ****************************************************************
ok: [158.252.83.114]

TASK [copy docker-compose file] ***************************************************************************
changed: [158.252.83.114]

TASK [deploy containers using docker-compose] *************************************************************
ok: [158.252.83.114]

PLAY RECAP ************************************************************************************************
158.252.83.114             : ok=32   changed=4    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ 
```

6. **Update security group** in modules/webserver/main.tf to add port 3000:

task9_sg_ingress_3000.png — web_sg snippet showing 3000/tcp ingress.

```
GNU nano 7.2                                   modules/webserver/main.tf *
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = [var.my_ip]
  }
  ingress {
    from_port   = 443
    to_port     = 443
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }
  ingress {
    from_port   = 80
    to_port     = 80
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }
ingress {
  from_port   = 3000
  to_port     = 3000
  protocol    = "tcp"
  cidr_blocks = ["0.0.0.0/0"]
}

  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
    prefix_list_ids = []
}
  tags = {
     Name = "${var.env_prefix}-default-sg"
```

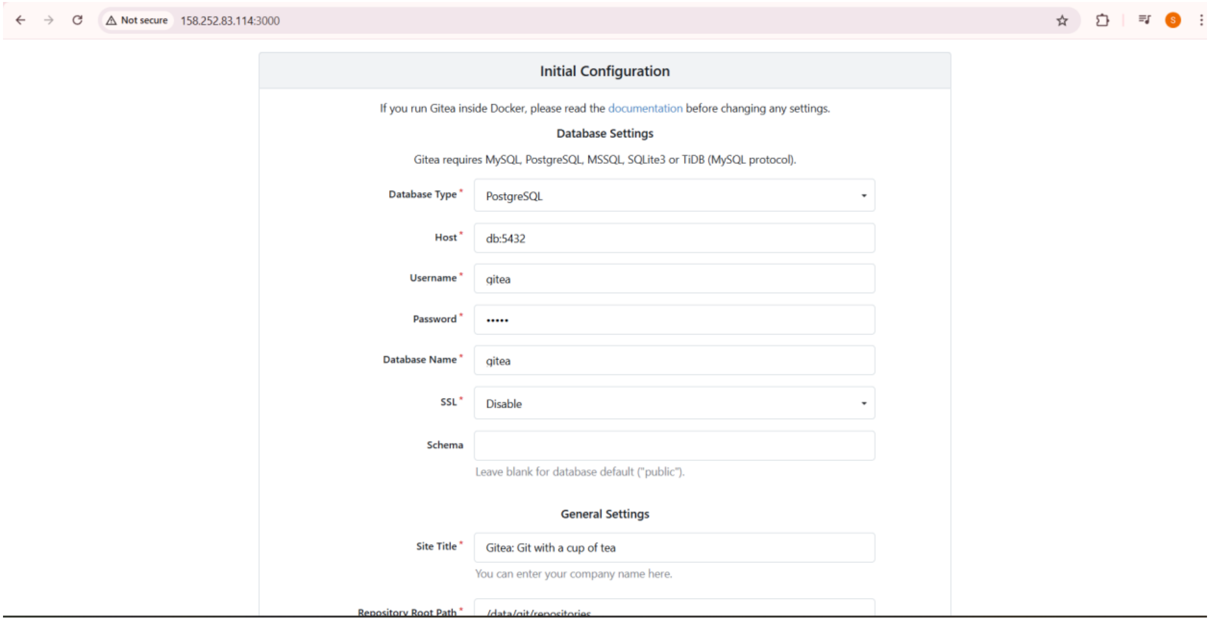
7. Apply:

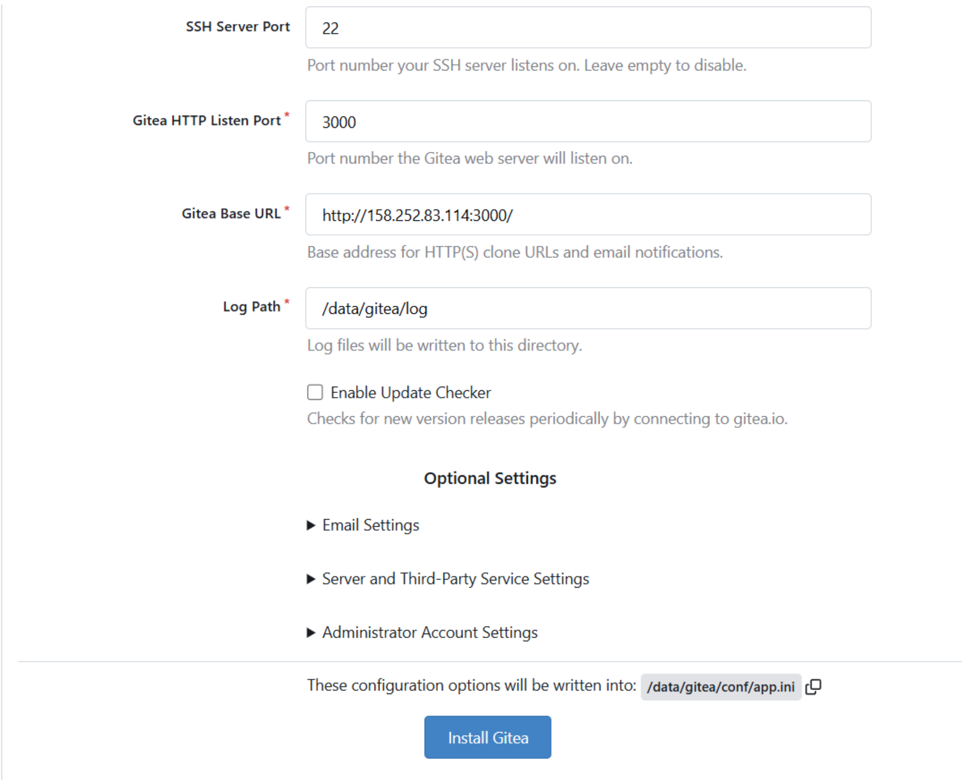
task9_terraform_apply_sg_3000.png — apply output updating SG.

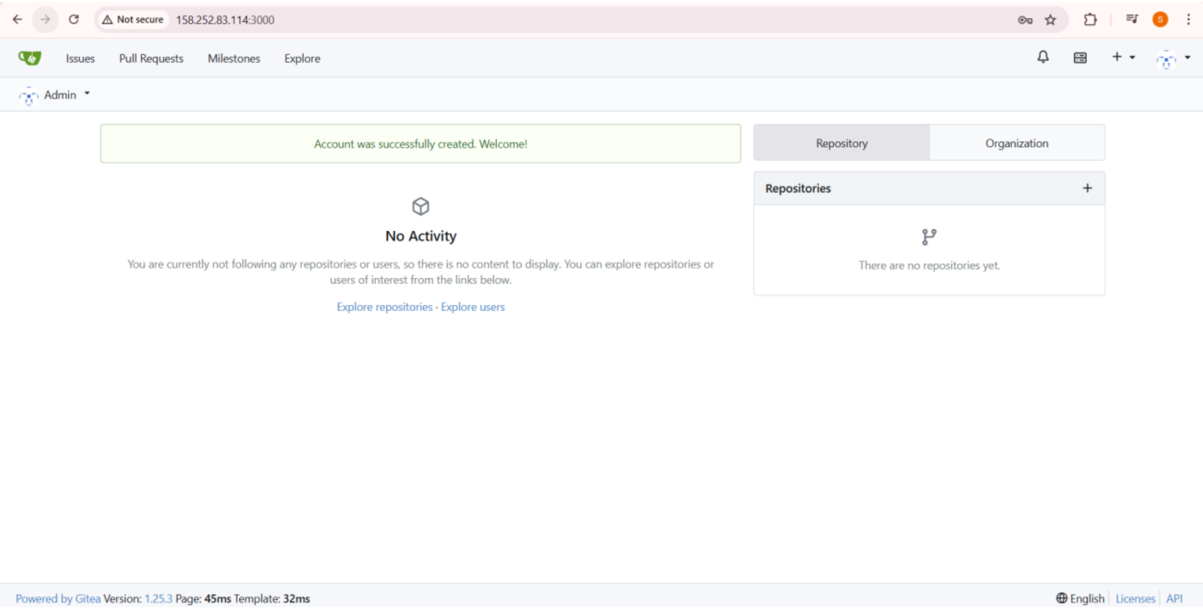
```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                bash  +  ···
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ terraform apply -auto-approve
                  + to_port          = 80
                },
              + {
                  + cidr_blocks      = [
                      + "4.240.39.195/32",
                    ]
                  + from_port        = 22
                  + ipv6_cidr_blocks = []
                  + prefix_list_ids  = []
                  + protocol         = "tcp"
                  + security_groups  = []
                  + self             = false
                  + to_port          = 22
                    # (1 unchanged attribute hidden)
                },
            ]
            name                   = "dev-web-sg-0"
            tags                   = {
                "Name" = "dev-default-sg"
            }
            # (9 unchanged attributes hidden)
        }

Plan: 0 to add, 1 to change, 0 to destroy.
module.myapp-webserver[0].aws_security_group.web_sg: Modifying... [id=sg-05d304624bf559a84]
module.myapp-webserver[0].aws_security_group.web_sg: Modifications complete after 1s [id=sg-05d304624bf559a84]

Apply complete! Resources: 0 added, 1 changed, 0 destroyed.

Outputs:

webserver_public_ips = [
  "158.252.83.114",
]
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $
```

8. Access Gitea:

task9_gitea_browser.png — browser showing Gitea UI.


Not secure 158.252.83.114:3000
Initial Configuration
If you run Gitea inside Docker, please read the documentation before changing any settings.
Database Settings
Gitea requires MySQL, PostgreSQL, MSSQL, SQLite3 or TiDB (MySQL protocol).
Database Type * PostgreSQL
Host * db:5432
Username * gitea
Password * •••••
Database Name * gitea
SSL * Disable
Schema
Leave blank for database default ("public").
General Settings
Site Title * Gitea: Git with a cup of tea
You can enter your company name here.
Repository Root Path * /data/git/repositories


SSH Server Port 22
Port number your SSH server listens on. Leave empty to disable.
Gitea HTTP Listen Port * 3000
Port number the Gitea web server will listen on.
Gitea Base URL * http://158.252.83.114:3000/
Base address for HTTP(S) clone URLs and email notifications.
Log Path * /data/gitea/log
Log files will be written to this directory.
Enable Update Checker
Checks for new version releases periodically by connecting to gitea.io.
Optional Settings
Email Settings
Server and Third-Party Service Settings
Administrator Account Settings
These configuration options will be written into: /data/gitea/conf/app.ini
Install Gitea



Task 10 – Automating Ansible with Terraform (null_resource)

Have Terraform trigger Ansible automatically after EC2 creation.

1. **Add null_resource** to main.tf:

task10_null_resource_main_tf.png — main.tf showing null_resource block.

```
  GNU nano 7.2                                   modules/webserver/main.tf *
resource "aws_instance" "myapp-server" {
  ami           = "ami-05524d6658fcf35b6" # Amazon Linux 2023 Kernel 6.1 AMI
  instance_type = var. instance_type
  subnet_id     = var.subnet_id
  vpc_security_group_ids = [aws_security_group.web_sg.id]
  availability_zone = var.availability_zone
  associate_public_ip_address = true
  key_name = aws_key_pair.ssh-key.key_name

  #user_data = file(var.script_path)

  tags = {
     Name = "${var.env_prefix}-ec2-instance-${var.instance_suffix}"
  }

}
resource "null_resource" "configure_server" {
  triggers = {
    webserver_public_ips_for_ansible = join(",", [for i in module.myapp-webserver : i.aws_instance.public_ip])
  }

  depends_on = [module.myapp-webserver]

  provisioner "local-exec" {
    command = <<-EOT
      ansible-playbook -i ${self.triggers.webserver_public_ips_for_ansible}, \
      --private-key "${var.private_key}" --user ec2-user \
      my-playbook.yaml
    EOT
  }
}
```

2. **Destroy and recreate infrastructure:**

task10_terraform_destroy_before_null.png — destroy output.

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                        bash

@23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform destroy -auto-approve
          - "Name" = "dev-default-sg"
        } -> null
      - tags_all               = {
          - "Name" = "dev-default-sg"
        } -> null
      - vpc_id                 = "vpc-0ade7c3c4a67a6421" -> null
        # (1 unchanged attribute hidden)
    }

Plan: 0 to add, 0 to change, 7 to destroy.

Changes to Outputs:
  - webserver_public_ips = [
      - "158.252.83.114",
    ] -> null
module.myapp-subnet.aws_default_route_table.main_rt: Destroying... [id=rtb-0b063f601309188d0]
module.myapp-webserver[0].aws_instance.myapp-server: Destroying... [id=i-0782572662a34a57c]
module.myapp-subnet.aws_default_route_table.main_rt: Destruction complete after 0s
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destroying... [id=igw-0cf955965564dac4c]
module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0782572662a34a57c, 00m10s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0cf955965564dac4c, 00m10s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destruction complete after 17s
module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0782572662a34a57c, 00m20s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Destruction complete after 20s
module.myapp-webserver[0].aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-0]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destroying... [id=subnet-0ccedd3e703d0f611]
module.myapp-webserver[0].aws_security_group.web_sg: Destroying... [id=sg-05d304624bf559a84]
module.myapp-webserver[0].aws_key_pair.ssh-key: Destruction complete after 1s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 1s
module.myapp-webserver[0].aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0ade7c3c4a67a6421]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 7 destroyed.
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

task10_terraform_apply_with_local_exec.png — apply output showing local-exec / Ansible execution.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform apply -auto-approve
data.http.my_ip: Read complete after 0s [id=https://icanhazip.com]
data.aws_subnet.default: Reading...
data.aws_vpc.default: Reading...
data.aws_subnet.default: Read complete after 0s [id=subnet-05ef1d5887d8aa227]
data.aws_vpc.default: Read complete after 0s [id=vpc-09f1466ca374f18e8]

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # module.myapp_webserver.aws_instance.myapp_server will be created
  + resource "aws_instance" "myapp_server" {
      + ami                                  = "ami-05524d6658fcf35b6"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = true
      + availability_zone                    = "ap-south-1a"
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t2.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = "dev-serverkey-0"
      + monitoring                           = (known after apply)
```

3. If Ansible fails due to readiness, add a wait play at the top of my-playbook.yaml:

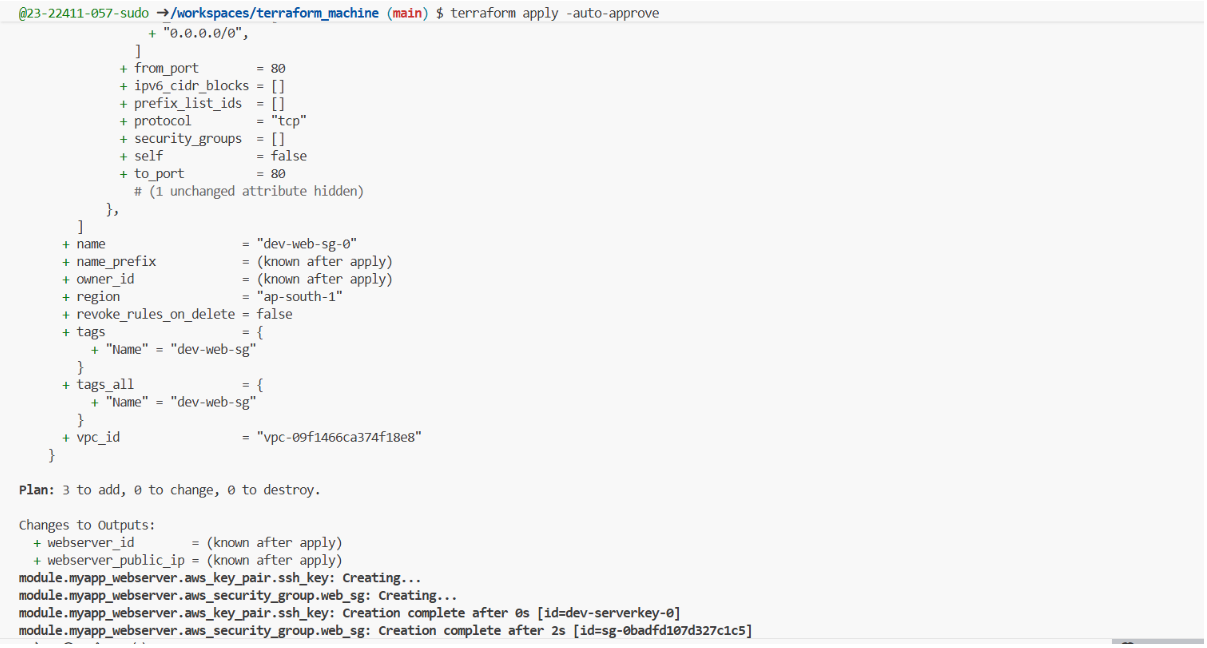
task10_my_playbook_wait_for_ssh.png — top wait play in file.

```
GNU nano 7.2                                                    my-playbook.yaml

- name: Wait for SSH to be available
  hosts: all
  gather_facts: no
  tasks:
    - name: Wait for port 22 to be ready
      wait_for:
        host: "{{ inventory_hostname }}"
        port: 22
        timeout: 300
        state: started

- name: Configure EC2 with Nginx and Gitea
  hosts: all
  become: true

  tasks:
    - name: Update packages
      yum:
        name: "*"
        state: latest

    - name: Install required packages
      yum:
        name:
          - nginx
          - git
        state: present

    - name: Enable and start Nginx
      service:
        name: nginx
```

4. **Destroy and apply again**:

task10_terraform_apply_after_wait.png — apply output after wait fix.

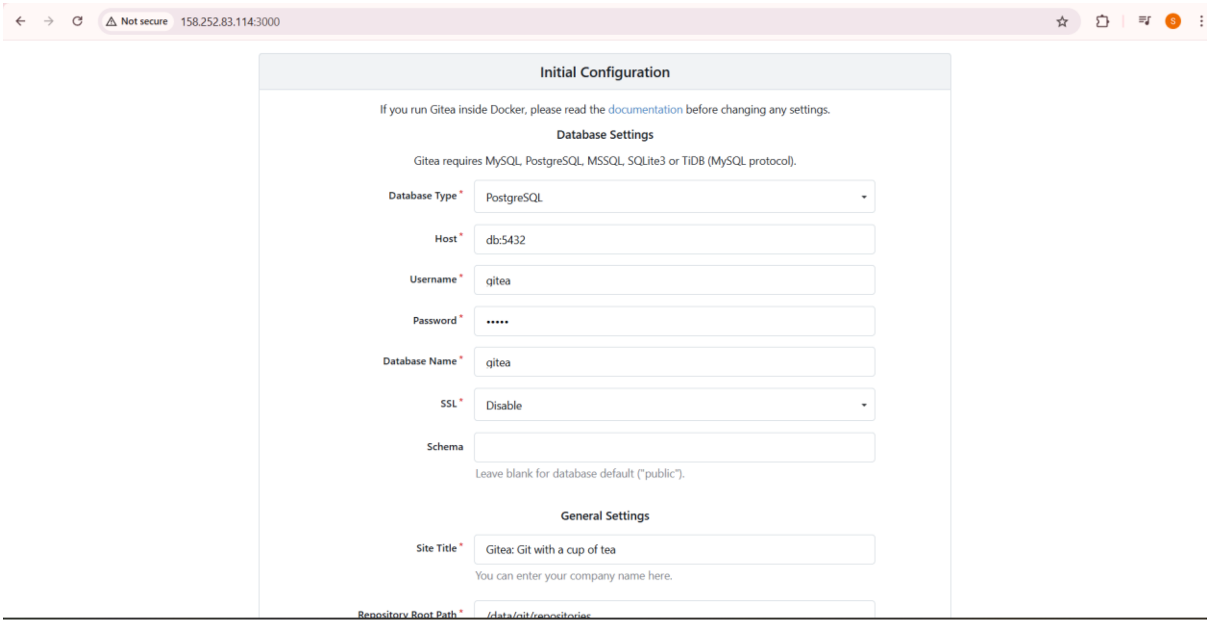
```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   bash

@23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform destroy -auto-approve
                - cidr_blocks      = [
                    - "0.0.0.0/0",
                  ]
                - from_port        = 80
                - ipv6_cidr_blocks = []
                - prefix_list_ids  = []
                - protocol         = "tcp"
                - security_groups  = []
                - self             = false
                - to_port          = 80
                  # (1 unchanged attribute hidden)
              },
          ] -> null
        - name                   = "dev-web-sg-0" -> null
        - owner_id               = "359416636630" -> null
        - region                 = "ap-south-1" -> null
        - revoke_rules_on_delete = false -> null
        - tags                   = {
            - "Name" = "dev-web-sg"
          } -> null
        - tags_all               = {
            - "Name" = "dev-web-sg"
          } -> null
        - vpc_id                 = "vpc-09f1466ca374f18e8" -> null
          # (1 unchanged attribute hidden)
    }

Plan: 0 to add, 0 to change, 2 to destroy.
module.myapp_webserver.aws_key_pair.ssh_key: Destroying... [id=dev-serverkey-0]
module.myapp_webserver.aws_security_group.web_sg: Destroying... [id=sg-0c6827bd5e2782815]
module.myapp_webserver.aws_key_pair.ssh_key: Destruction complete after 0s
module.myapp_webserver.aws_security_group.web_sg: Destruction complete after 0s

Destroy complete! Resources: 2 destroyed.
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```



```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform apply -auto-approve
                  + "0.0.0.0/0",
                ]
              + from_port        = 80
              + ipv6_cidr_blocks = []
              + prefix_list_ids  = []
              + protocol         = "tcp"
              + security_groups  = []
              + self             = false
              + to_port          = 80
                # (1 unchanged attribute hidden)
            },
        ]
      + name                   = "dev-web-sg-0"
      + name_prefix            = (known after apply)
      + owner_id               = (known after apply)
      + region                 = "ap-south-1"
      + revoke_rules_on_delete = false
      + tags                   = {
          + "Name" = "dev-web-sg"
        }
      + tags_all               = {
          + "Name" = "dev-web-sg"
        }
      + vpc_id                 = "vpc-09f1466ca374f18e8"
    }

Plan: 3 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + webserver_id        = (known after apply)
  + webserver_public_ip = (known after apply)
module.myapp_webserver.aws_key_pair.ssh_key: Creating...
module.myapp_webserver.aws_security_group.web_sg: Creating...
module.myapp_webserver.aws_key_pair.ssh_key: Creation complete after 0s [id=dev-serverkey-0]
module.myapp_webserver.aws_security_group.web_sg: Creation complete after 2s [id=sg-0badfd107d327c1c5]
```

5. Verify Gitea / Nginx application is reachable at the appropriate URL.

task10_app_browser_post_null_resource.png — browser confirming app is up after Terraform+Ansible.



Task 11 – Dynamic inventory with aws_ec2 plugin

Let Ansible discover EC2 instances dynamically via the aws_ec2 inventory plugin.

1. Update ansible.cfg:

task11_ansible_cfg_aws_ec2.png — content of ansible.cfg with plugin configuration.

```
GNU nano 7.2                                                    ansible.cfg
[defaults]
host_key_checking = False
interpreter_python = /usr/bin/python3
deprecation_warnings = False

enable_plugins = aws_ec2
private_key_file = ~/.ssh/id_ed25519
```

2. **Create inventory_aws_ec2.yaml**:

task11_inventory_aws_ec2_created.png — file created.

```
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ touch inventory_aws_ec2.yaml
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ ls -la inventory_aws_ec2.yaml
-rw-rw-rw- 1 codespace codespace 0 Jan 11 08:21 inventory_aws_ec2.yaml
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $
```

task11_inventory_aws_ec2_initial.png — content of initial inventory file.

```
GNU nano 7.2                                          inventory_aws_ec2.yaml *
---
plugin: aws_ec2
regions:
  - me-central-1
```

3. **Ensure Terraform code** includes both dev and prod webservers in **main.tf**:

task11_main_tf_dev_prod_modules.png — snippet of main.tf showing myapp-webserver and myapp-webserver-prod.

```
GNU nano 7.2                                          main.tf *
provider "aws" {
  region                   = "me-central-1"
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_vpc" "myapp_vpc" {
  cidr_block           = var.vpc_cidr_block
  enable_dns_hostnames = true
  tags = {
    Name = "${var.env_prefix}-vpc"
  }
}

module "myapp-subnet" {
  source                  = "./modules/subnet"
  vpc_id                  = aws_vpc.myapp_vpc.id
  subnet_cidr_block       = var.subnet_cidr_block
  availability_zone       = var.availability_zone
  env_prefix              = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}

# DEV Webserver
module "myapp-webserver" {
  source          = "./modules/webserver"
  env_prefix      = var.env_prefix
```

```
  GNU nano 7.2                                                              main.tf *
  env_prefix         = var.env_prefix
  instance_type      = var.instance_type
  availability_zone  = var.availability_zone
  public_key         = var.public_key
  my_ip              = local.my_ip
  vpc_id             = aws_vpc.myapp_vpc.id
  subnet_id          = module.myapp-subnet.subnet.id

  count             = 1
  instance_suffix   = count.index
}

# PROD Webserver
module "myapp-webserver-prod" {
  source             = "./modules/webserver"
  env_prefix         = "prod"
  instance_type      = "t3.nano"
  availability_zone  = var.availability_zone
  public_key         = var.public_key
  my_ip              = local.my_ip
  vpc_id             = aws_vpc.myapp_vpc.id
  subnet_id          = module.myapp-subnet.subnet.id

  count             = 1
  instance_suffix   = count.index
}
```

4. **Add outputs in outputs.tf**:

task11_outputs_tf_dev_prod_ips.png — content of outputs.tf with both outputs.

```
  GNU nano 7.2                                                  outputs.tf
output "webserver_public_ips" {
  value = [for i in module.myapp-webserver : i.aws_instance.public_ip]
}

output "prod-webserver_public_ips" {
  value = [for i in module.myapp-webserver-prod : i.aws_instance.public_ip]
}
```

5. **Rebuild infra**:

task11_terraform_apply_dynamic_setup.png — apply output for dev/prod instances.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform init
Initializing the backend...
Initializing modules...
- myapp-subnet in modules/subnet
- myapp-webserver in modules/webserver
- myapp-webserver-prod in modules/webserver
Initializing provider plugins...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/null from the dependency lock file
- Reusing previous version of hashicorp/http from the dependency lock file
- Using previously-installed hashicorp/aws v6.28.0
- Using previously-installed hashicorp/null v3.2.4
- Using previously-installed hashicorp/http v3.5.0

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

```
@23-22411-061-rgb →/workspaces/terraform_machine (main) $ terraform apply -auto-approve
    }

Plan: 3 to add, 1 to change, 1 to destroy.

Changes to Outputs:
  + prod-webserver_public_ips = [
      + (known after apply),
    ]
null_resource.configure_server: Destroying... [id=1690333078838817863]
null_resource.configure_server: Destruction complete after 0s
module.myapp-webserver-prod[0].aws_key_pair.ssh-key: Creating...
module.myapp-webserver-prod[0].aws_security_group.web_sg: Creating...
module.myapp-webserver[0].aws_security_group.web_sg: Modifying... [id=sg-0863b9335d553f9c5]
module.myapp-webserver-prod[0].aws_key_pair.ssh-key: Creation complete after 1s [id=prod-serverkey-0]
module.myapp-webserver[0].aws_security_group.web_sg: Modifications complete after 2s [id=sg-0863b9335d553f9c5]
module.myapp-webserver-prod[0].aws_security_group.web_sg: Creation complete after 3s [id=sg-0c546cd1ddd0cb2b9]
module.myapp-webserver-prod[0].aws_instance.myapp-server: Creating...
module.myapp-webserver-prod[0].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver-prod[0].aws_instance.myapp-server: Creation complete after 13s [id=i-02dd29264b6acb6aa]

Apply complete! Resources: 3 added, 1 changed, 1 destroyed.

Outputs:

prod-webserver_public_ips = [
  "3.29.15.146",
]
webserver_public_ips = [
  "51.112.187.163",
]
@23-22411-061-rgb →/workspaces/terraform_machine (main) $
```

task11_terraform_output_dynamic_ips.png — terraform output showing dev and prod IPs.

```
]
@23-22411-061-rgb →/workspaces/terraform_machine (main) $ terraform output
prod-webserver_public_ips = [
  "3.29.15.146",
]
webserver_public_ips = [
  "51.112.187.163",
]
```

6. **Install boto3 and botocore**:

task11_boto_install.png — pip install output.

```
@23-22411-057-sudo ➜ /workspaces/terraform_machine (main) $ $(which python3) -m pip install --user boto3 botocore
Collecting boto3
  Downloading boto3-1.42.25-py3-none-any.whl.metadata (6.8 kB)
Collecting botocore
  Downloading botocore-1.42.25-py3-none-any.whl.metadata (5.9 kB)
Collecting jmespath<2.0.0,>=0.7.1 (from boto3)
  Downloading jmespath-1.0.1-py3-none-any.whl.metadata (7.6 kB)
Collecting s3transfer<0.17.0,>=0.16.0 (from boto3)
  Downloading s3transfer-0.16.0-py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in /home/codespace/.local/lib/python3.12/site-packages (from botocore) (2.9.0.post0)
Requirement already satisfied: urllib3!=2.2.0,<3,>=1.25.4 in /home/codespace/.local/lib/python3.12/site-packages (from botocore) (2.5.0)
Requirement already satisfied: six>=1.5 in /home/codespace/.local/lib/python3.12/site-packages (from python-dateutil<3.0.0,>=2.1->botocore) (1.17.0)
Downloading boto3-1.42.25-py3-none-any.whl (140 kB)
Downloading botocore-1.42.25-py3-none-any.whl (14.6 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 14.6/14.6 MB 29.2 MB/s  0:00:00
Downloading jmespath-1.0.1-py3-none-any.whl (20 kB)
Downloading s3transfer-0.16.0-py3-none-any.whl (86 kB)
Installing collected packages: jmespath, botocore, s3transfer, boto3
Successfully installed boto3-1.42.25 botocore-1.42.25 jmespath-1.0.1 s3transfer-0.16.0
@23-22411-057-sudo ➜ /workspaces/terraform_machine (main) $
```

task11_boto_version.png — printed version line.

```
@23-22411-057-sudo ➜ /workspaces/terraform_machine (main) $ $(which python3) -c "import boto3, botocore; print('boto3:', boto3.__version__, 'botocore:', botocore.__version__)"
boto3: 1.42.25 botocore: 1.42.25
@23-22411-057-sudo ➜ /workspaces/terraform_machine (main) $
```

7. **Check inventory graph**:

task11_ansible_inventory_graph_initial.png — graph output showing @all, @ungrouped, and @aws_ec2 hosts.

```
@23-22411-057-sudo ➜ /workspaces/terraform_machine (main) $ ansible-inventory -i inventory_aws_ec2.yaml --graph
Failed to parse inventory with 'auto' plugin.

<<< caused by >>>

inventory config '/workspaces/terraform_machine/inventory_aws_ec2.yaml' specifies unknown plugin 'aws_ec2'
Origin: <inventory plugin 'auto' with source '/workspaces/terraform_machine/inventory_aws_ec2.yaml'>

[WARNING]: Failed to parse inventory with 'yaml' plugin: Plugin configuration YAML file, not YAML inventory

Failed to parse inventory with 'yaml' plugin.

<<< caused by >>>

Plugin configuration YAML file, not YAML inventory
Origin: <inventory plugin 'yaml' with source '/workspaces/terraform_machine/inventory_aws_ec2.yaml'>

[WARNING]: Failed to parse inventory with 'ini' plugin: Failed to parse inventory: Invalid host pattern '---' supplied, '---' is normally a sign this is a YAML file.

Failed to parse inventory with 'ini' plugin.

<<< caused by >>>

Failed to parse inventory: Invalid host pattern '---' supplied, '---' is normally a sign this is a YAML file.
Origin: /workspaces/terraform_machine/inventory_aws_ec2.yaml

[WARNING]: Unable to parse /workspaces/terraform_machine/inventory_aws_ec2.yaml as an inventory source
[WARNING]: No inventory was parsed, only implicit localhost is available
@all:
  |--@ungrouped:
@23-22411-057-sudo ➜ /workspaces/terraform_machine (main) $
```

Task 12 – Filtering EC2 instances by tags & instance type

Augment the inventory plugin to group by tags and instance type, then limit plays to specific groups.

1. **Modify inventory_aws_ec2.yaml** to add tag-based grouping:

task12_inventory_aws_ec2_tag_groups.png — updated inventory file.

```
GNU nano 7.2                                         inventory_aws_ec2.yaml *
---
plugin: aws_ec2
regions:
  - me-central-1

keyed_groups:
  - key: tags
    prefix: tag
    separator: "_"
```

task12_inventory_graph_tag_groups.png — ansible-inventory graph showing tag groups.\

```
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ ansible-inventory -i inventory_aws_ec2.yaml --graph
Failed to parse inventory with 'auto' plugin.

<<< caused by >>>

inventory config '/workspaces/terraform_machine/inventory_aws_ec2.yaml' specifies unknown plugin 'aws_ec2'
Origin: <inventory plugin 'auto' with source '/workspaces/terraform_machine/inventory_aws_ec2.yaml'>

[WARNING]: Failed to parse inventory with 'yaml' plugin: Plugin configuration YAML file, not YAML inventory

Failed to parse inventory with 'yaml' plugin.

<<< caused by >>>

Plugin configuration YAML file, not YAML inventory
Origin: <inventory plugin 'yaml' with source '/workspaces/terraform_machine/inventory_aws_ec2.yaml'>

[WARNING]: Failed to parse inventory with 'ini' plugin: Failed to parse inventory: Invalid host pattern '---' supplied, '---' is normally a sign this is a YAML file.

Failed to parse inventory with 'ini' plugin.

<<< caused by >>>

Failed to parse inventory: Invalid host pattern '---' supplied, '---' is normally a sign this is a YAML file.
Origin: /workspaces/terraform_machine/inventory_aws_ec2.yaml

[WARNING]: Unable to parse /workspaces/terraform_machine/inventory_aws_ec2.yaml as an inventory source
[WARNING]: No inventory was parsed, only implicit localhost is available
@all:
  |--@ungrouped:
○ @23-22411-057-sudo ➜/workspaces/terraform_machine (main) $
```

2. **Extend to group by instance type as well**:

task12_inventory_aws_ec2_instance_type_groups.png — inventory with tags + instance_type.

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

GNU nano 7.2                                         inventory_aws_ec2.yaml *
---
plugin: aws_ec2
regions:
  - me-central-1

keyed_groups:
  - key: tags
    prefix: tag
    separator: "_"

 - key: instance_type
    prefix: instance_type
    separator: "_"
```

task12_inventory_graph_full.png — graph showing both tag and instance type groups.

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ ansible-inventory -i inventory_aws_ec2.yaml --graph
Origin: /workspaces/terraform_machine/inventory_aws_ec2.yaml:11:2

 9     separator: "_"
10
11  - key: instance_type
    ^ column 2

[WARNING]: Failed to parse inventory with 'yaml' plugin: YAML parsing failed: While parsing a block mapping did not find expected key.

Failed to parse inventory with 'yaml' plugin.

<<< caused by >>>

YAML parsing failed: While parsing a block mapping did not find expected key.
Origin: <inventory plugin 'yaml' with source '/workspaces/terraform_machine/inventory_aws_ec2.yaml'>

[WARNING]: Failed to parse inventory with 'ini' plugin: Failed to parse inventory: Invalid host pattern '---' supplied, '---' is normally a sign this is a YAML file.

Failed to parse inventory with 'ini' plugin.

<<< caused by >>>

Failed to parse inventory: Invalid host pattern '---' supplied, '---' is normally a sign this is a YAML file.
Origin: /workspaces/terraform_machine/inventory_aws_ec2.yaml

[WARNING]: Unable to parse /workspaces/terraform_machine/inventory_aws_ec2.yaml as an inventory source
[WARNING]: No inventory was parsed, only implicit localhost is available
@all:
  |--@ungrouped:
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

3. **Prepare my-playbook.yaml** for nginx+SSL+PHP on hosts: all:

task12_my_playbook_all_hosts.png — playbook configured for hosts: all.

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                  nano

 GNU nano 7.2                              my-playbook.yaml *
---
- name: Configure nginx web server
  hosts: all
  become: true
  tasks:
    - name: install nginx and update cache
      yum:
        name: nginx
        state: present
        update_cache: yes

    - name: install openssl
      yum:
        name: openssl
        state: present

    - name: start nginx server
      service:
        name: nginx
        state: started
        enabled: true

- name: Configure SSL certificates
  hosts: all
  become: true
  tasks:
    - name: Create SSL private directory
```

```
 GNU nano 7.2                              my-playbook.yaml *
        headers:
          X-aws-ec2-metadata-token-ttl-seconds: "3600"
        return_content: yes
      register: imdsv2_token

    - name: Get current public IP
      uri:
        url: "http://169.254.169.254/latest/meta-data/public-ipv4"
        headers:
          X-aws-ec2-metadata-token: "{{ imdsv2_token.content }}"
        return_content: yes
      register: public_ip

    - name: Show current public IP
      debug:
        msg: "Public IP: {{ public_ip.content }}"

    - name: Save public IP as fact
      set_fact:
        server_public_ip: "{{ public_ip.content }}"

    - name: Generate self-signed SSL certificate
      command: >
        openssl req -x509 -nodes -days 365
        -newkey rsa:2048
        -keyout /etc/ssl/private/selfsigned.key
        -out /etc/ssl/certs/selfsigned.crt
```

4. **Run on all instances**:

task12_ansible_play_all.png — output showing all dev+prod hosts configured.

```
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ ansible-playbook -i inventory_aws_ec2.yaml my-playbook.yaml
<<< caused by >>>

YAML parsing failed: While parsing a block mapping did not find expected key.
Origin: <inventory plugin 'yaml' with source '/workspaces/terraform_machine/inventory_aws_ec2.yaml'>

[WARNING]: Failed to parse inventory with 'ini' plugin: Failed to parse inventory: Invalid host pattern '---' supplied, '---' is normally a sign this is a YAML file.

Failed to parse inventory with 'ini' plugin.

<<< caused by >>>

Failed to parse inventory: Invalid host pattern '---' supplied, '---' is normally a sign this is a YAML file.
Origin: /workspaces/terraform_machine/inventory_aws_ec2.yaml

[WARNING]: Unable to parse /workspaces/terraform_machine/inventory_aws_ec2.yaml as an inventory source
[WARNING]: No inventory was parsed, only implicit localhost is available
[WARNING]: provided hosts list is empty, only localhost is available. Note that the implicit localhost does not match 'all'

PLAY [Configure nginx web server] ******************************************************************************************
skipping: no hosts matched

PLAY [Configure SSL certificates] ******************************************************************************************
skipping: no hosts matched

PLAY [Deploy Nginx website and configuration files] ************************************************************************
skipping: no hosts matched

PLAY RECAP *****************************************************************************************************************

@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $
```

5. **Run only on dev instances**:

task12_ansible_play_dev_only.png — filtered run for dev group.

```
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ ansible-playbook -i inventory_aws_ec2.yaml -l tag_Name_dev_* my-playbook.yaml

YAML parsing failed: While parsing a block mapping did not find expected key.
Origin: <inventory plugin 'yaml' with source '/workspaces/terraform_machine/inventory_aws_ec2.yaml'>

[WARNING]: Failed to parse inventory with 'ini' plugin: Failed to parse inventory: Invalid host pattern '---' supplied, '---' is normally a sign this is a YAML file.

Failed to parse inventory with 'ini' plugin.

<<< caused by >>>

Failed to parse inventory: Invalid host pattern '---' supplied, '---' is normally a sign this is a YAML file.
Origin: /workspaces/terraform_machine/inventory_aws_ec2.yaml

[WARNING]: Unable to parse /workspaces/terraform_machine/inventory_aws_ec2.yaml as an inventory source
[WARNING]: No inventory was parsed, only implicit localhost is available
[WARNING]: provided hosts list is empty, only localhost is available. Note that the implicit localhost does not match 'all'
[WARNING]: Could not match supplied host pattern, ignoring: tag_Name_dev_*

PLAY [Configure nginx web server] ******************************************************************************************
skipping: no hosts matched

PLAY [Configure SSL certificates] ******************************************************************************************
skipping: no hosts matched

PLAY [Deploy Nginx website and configuration files] ************************************************************************
skipping: no hosts matched

PLAY RECAP *****************************************************************************************************************

@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $
```

6. **Run only on prod instances**:

task12_ansible_play_prod_only.png — filtered run for prod group.

```
@23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ ansible-playbook -i inventory_aws_ec2.yaml -l tag_Name_prod_* my-playbook.yaml

YAML parsing failed: While parsing a block mapping did not find expected key.
Origin: <inventory plugin 'yaml' with source '/workspaces/terraform_machine/inventory_aws_ec2.yaml'>

[WARNING]: Failed to parse inventory with 'ini' plugin: Failed to parse inventory: Invalid host pattern '---' supplied, '---' is normally a sign this is a YAML file.

Failed to parse inventory with 'ini' plugin.

<<< caused by >>>

Failed to parse inventory: Invalid host pattern '---' supplied, '---' is normally a sign this is a YAML file.
Origin: /workspaces/terraform_machine/inventory_aws_ec2.yaml

[WARNING]: Unable to parse /workspaces/terraform_machine/inventory_aws_ec2.yaml as an inventory source
[WARNING]: No inventory was parsed, only implicit localhost is available
[WARNING]: provided hosts list is empty, only localhost is available. Note that the implicit localhost does not match 'all'
[WARNING]: Could not match supplied host pattern, ignoring: tag_Name_prod_*

PLAY [Configure nginx web server] ******************************************************************************************
skipping: no hosts matched

PLAY [Configure SSL certificates] ******************************************************************************************
skipping: no hosts matched

PLAY [Deploy Nginx website and configuration files] ************************************************************************
skipping: no hosts matched

PLAY RECAP *****************************************************************************************************************
```

7. **Run only on t3.micro** instances:

task12_ansible_play_t3_micro.png — run for instance_type_t3_micro.

```
@23-22411-057-sudo ➜ /workspaces/terraform_machine (main) $ ansible-playbook -i inventory_aws_ec2.yaml -l instance_type_t3_micro my-playbook.yaml

YAML parsing failed: While parsing a block mapping did not find expected key.
Origin: <inventory plugin 'yaml' with source '/workspaces/terraform_machine/inventory_aws_ec2.yaml'>

[WARNING]: Failed to parse inventory with 'ini' plugin: Failed to parse inventory: Invalid host pattern '---' supplied, '---' is normally a sign this is a YAML file.

Failed to parse inventory with 'ini' plugin.

<<< caused by >>>

Failed to parse inventory: Invalid host pattern '---' supplied, '---' is normally a sign this is a YAML file.
Origin: /workspaces/terraform_machine/inventory_aws_ec2.yaml

[WARNING]: Unable to parse /workspaces/terraform_machine/inventory_aws_ec2.yaml as an inventory source
[WARNING]: No inventory was parsed, only implicit localhost is available
[WARNING]: provided hosts list is empty, only localhost is available. Note that the implicit localhost does not match 'all'
[WARNING]: Could not match supplied host pattern, ignoring: instance_type_t3_micro

PLAY [Configure nginx web server] ******************************************************************************************
skipping: no hosts matched

PLAY [Configure SSL certificates] ******************************************************************************************
skipping: no hosts matched

PLAY [Deploy Nginx website and configuration files] ************************************************************************
skipping: no hosts matched

PLAY RECAP *****************************************************************************************************************

@23-22411-057-sudo ➜ /workspaces/terraform_machine (main) $
```

8. **Run only on t3.nano** instances:

task12_ansible_play_t3_nano.png — run for instance_type_t3_nano.

```
@23-22411-057-sudo ➜ /workspaces/terraform_machine (main) $ ansible-playbook -i inventory_aws_ec2.yaml -l instance_type_t3_nano my-playbook.yaml

YAML parsing failed: While parsing a block mapping did not find expected key.
Origin: <inventory plugin 'yaml' with source '/workspaces/terraform_machine/inventory_aws_ec2.yaml'>

[WARNING]: Failed to parse inventory with 'ini' plugin: Failed to parse inventory: Invalid host pattern '---' supplied, '---' is normally a sign this is a YAML file.

Failed to parse inventory with 'ini' plugin.

<<< caused by >>>

Failed to parse inventory: Invalid host pattern '---' supplied, '---' is normally a sign this is a YAML file.
Origin: /workspaces/terraform_machine/inventory_aws_ec2.yaml

[WARNING]: Unable to parse /workspaces/terraform_machine/inventory_aws_ec2.yaml as an inventory source
[WARNING]: No inventory was parsed, only implicit localhost is available
[WARNING]: provided hosts list is empty, only localhost is available. Note that the implicit localhost does not match 'all'
[WARNING]: Could not match supplied host pattern, ignoring: instance_type_t3_nano

PLAY [Configure nginx web server] ******************************************************************************************
skipping: no hosts matched

PLAY [Configure SSL certificates] ******************************************************************************************
skipping: no hosts matched

PLAY [Deploy Nginx website and configuration files] ************************************************************************
skipping: no hosts matched

PLAY RECAP *****************************************************************************************************************

@23-22411-057-sudo ➜ /workspaces/terraform_machine (main) $
```

9. **Update ansible.cfg** to use inventory by default:

task12_ansible_cfg_inventory_default.png — ansible.cfg with inventory = ./inventory_aws_ec2.yaml.

```
 GNU nano 7.2                                                                    ansible.cfg *
[defaults]
host_key_checking = False
interpreter_python = /usr/bin/python3
deprecation_warnings = False

enable_plugins = aws_ec2
private_key_file = ~/.ssh/id_ed25519

[defaults]
inventory = ./inventory_aws_ec2.yaml
```

10. Now you can simply run:

task12_ansible_play_t3_nano_no_i.png — run using default inventory (no -i).

```
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ ansible-playbook -l instance_type_t3_nano my-playbook.yaml
[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an ansible.cfg source. For more information see https://docs.an
.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writable-dir
[WARNING]: No inventory was parsed, only implicit localhost is available
[WARNING]: provided hosts list is empty, only localhost is available. Note that the implicit localhost does not match 'all'
[WARNING]: Could not match supplied host pattern, ignoring: instance_type_t3_nano

PLAY [Configure nginx web server] ************************************************************************************************************************
skipping: no hosts matched

PLAY [Configure SSL certificates] ************************************************************************************************************************
skipping: no hosts matched

PLAY [Deploy Nginx website and configuration files] ******************************************************************************************************
skipping: no hosts matched

PLAY RECAP ***********************************************************************************************************************************************

@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

Task 13 – Ansible roles: nginx, ssl, webapp

Reorganize your configuration into **roles**.

1. **Update main.tf** for a simple dev environment with 1 instance (as shown previously).

task13_main_tf_single_dev.png — main.tf snippet with single myapp-webserver count=1.

```
  vpc_id = aws_vpc.myapp_vpc.id
  subnet_cidr_block = var.subnet_cidr_block
  availability_zone = var.availability_zone
  env_prefix = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}

module "myapp-webserver" {
  source            = "./modules/webserver"
  env_prefix        = var.env_prefix
  instance_type     = var.instance_type
  availability_zone = var.availability_zone
  public_key        = var.public_key
  my_ip             = local.my_ip
  vpc_id            = aws_vpc.myapp_vpc.id
  subnet_id         = module.myapp-subnet.subnet.id

  # Loop count
  count           = 1
  # Use count.index to differentiate instances
  instance_suffix   = count.index
}
~
~
~
~
~
~
-- INSERT --
0   0
```

2. **Create /ansible structure**:

task13_ansible_structure_created.png — ls -R showing ansible/, inventory/, roles/, ansible.cfg, my-playbook.yaml.

```
● @23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ nano main.tf
● @23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ mkdir -p ansible/{inventory,roles}
● @23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ touch ansible/ansible.cfg ansible/my-playbook.yaml
● @23-22411-057-sudo ➜/workspaces/terraform_machine (main) $ ls -R ansible
ansible:
ansible.cfg  inventory  my-playbook.yaml  roles

ansible/inventory:

ansible/roles:
○ @23-22411-057-sudo ➜/workspaces/terraform_machine (main) $
```

3. **ansible/ansible.cfg**:

task13_ansible_cfg_project.png — content of ansible/ansible.cfg.

```
 GNU nano 7.2                                                    ansible/ansible.cfg *
[defaults]
host_key_checking=False
interpreter_python = /usr/bin/python3
```

4. **ansible/inventory/hosts**:

task13_ansible_inventory_hosts.png — content of ansible/inventory/hosts.

```
[defaults]
host_key_checking=False
interpreter_python = /usr/bin/python3
~
~
~
~
~
~
~
~
~
~
~
```

5. **Create roles**:

task13_roles_created.png — ls -R under ansible/roles showing nginx, ssl, webapp.

```
● @23-22411-057-sudo →/workspaces/terraform_machine (main) $ cd ansible/roles
● @23-22411-057-sudo →/workspaces/terraform_machine/ansible/roles (main) $ ansible-galaxy role init nginx
  ansible-galaxy role init ssl
  ansible-galaxy role init webapp
  - Role nginx was created successfully
  - Role ssl was created successfully
  - Role webapp was created successfully
● @23-22411-057-sudo →/workspaces/terraform_machine/ansible/roles (main) $ cd ..
● @23-22411-057-sudo →/workspaces/terraform_machine/ansible (main) $ ls -R roles
  roles:
  nginx  ssl  webapp

  roles/nginx:
  README.md  defaults  files  handlers  meta  tasks  templates  tests  vars

  roles/nginx/defaults:
  main.yml

  roles/nginx/files:

  roles/nginx/handlers:
  main.yml

  roles/nginx/meta:
  main.yml

  roles/nginx/tasks:
  main.yml

  roles/nginx/templates:
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                  bash - ansible

@23-22411-057-sudo →/workspaces/terraform_machine/ansible (main) $ ls -R roles
inventory  test.yml

roles/ssl/vars:
main.yml

roles/webapp:
README.md  defaults  files  handlers  meta  tasks  templates  tests  vars

roles/webapp/defaults:
main.yml

roles/webapp/files:

roles/webapp/handlers:
main.yml

roles/webapp/meta:
main.yml

roles/webapp/tasks:
main.yml

roles/webapp/templates:

roles/webapp/tests:
inventory  test.yml

roles/webapp/vars:
main.yml
@23-22411-057-sudo →/workspaces/terraform_machine/ansible (main) $
```

6. **Role: nginx**

task13_nginx_handlers_main.png — content of roles/nginx/handlers/main.yml.

```
#SPDX-License-Identifier: MIT-0
# handlers file for nginx

- name: Restart nginx
  service:
    name: nginx
    state: restarted
~
~
~
~
~
~
~
~
~
~
~
```

task13_nginx_tasks_main.png — content of roles/nginx/tasks/main.yml.

```
#SPDX-License-Identifier: MIT-0
# tasks file for nginx

- name: Install nginx
  yum:
    name: nginx
    state: present
    update_cache: yes
  notify: Restart nginx

- name: Install openssl
  yum:
    name: openssl
    state: present

- name: Start and enable nginx
  service:
    name: nginx
    state: started
    enabled: true
~
~
~
```

7. **First role-based playbook** ansible/my-playbook.yaml:

task13_my_playbook_nginx_only.png — playbook using only nginx role.

```
---
- name: Deploy NGINX Web Stack with SSL and PHP
  hosts: nginx
  become: true
  roles:
    - nginx
~
~
~
~
~
~
~
~
~
```

task13_ansible_play_nginx_only.png — playbook output.

```
@23-22411-061-rgb →/workspaces/terraform_machine/ansible (main) $ ansible-playbook -i inventory/hosts my-playbook.yam
l

PLAY [Deploy NGINX Web Stack with SSL and PHP] ************************************************************************

TASK [Gathering Facts] ************************************************************************************************
ok: [51.112.187.163]

TASK [nginx : Install nginx] ******************************************************************************************
changed: [51.112.187.163]

TASK [nginx : Install openssl] ****************************************************************************************
ok: [51.112.187.163]

TASK [nginx : Start and enable nginx] *********************************************************************************
changed: [51.112.187.163]

RUNNING HANDLER [nginx : Restart nginx] *******************************************************************************
changed: [51.112.187.163]

PLAY RECAP ************************************************************************************************************
51.112.187.163             : ok=5    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

@23-22411-061-rgb →/workspaces/terraform_machine/ansible (main) $
```

task13_nginx_browser_roles.png — browser showing nginx default page via roles.

8. Role: ssl

task13_ssl_defaults_main.png — content of roles/ssl/defaults/main.yml.

```
#SPDX-License-Identifier: MIT-0

# defaults file for ssl
imdsv2_token_ttl: "3600"
ssl_days_valid: 365
~
~
~
~
~
~
~
~
~
~
```

task13_ssl_tasks_main.png — content of roles/ssl/tasks/main.yml.

```
    path: /etc/ssl/certs
    state: directory
    mode: '0755'

- name: Get IMDSv2 token
  uri:
    url: http://169.254.169.254/latest/api/token
    method: PUT
    headers:
      X-aws-ec2-metadata-token-ttl-seconds: "{{ imdsv2_token_ttl }}"
    return_content: yes
  register: imds_token

- name: Get public IP
  uri:
    url: http://169.254.169.254/latest/meta-data/public-ipv4
    headers:
      X-aws-ec2-metadata-token: "{{ imds_token.content }}"
    return_content: yes
  register: public_ip

- name: Save public IP as fact
  set_fact:
    server_public_ip: "{{ public_ip.content }}"

- name: Generate self-signed certificate
  command: >
    openssl req -x509 -nodes -days {{ ssl_days_valid }}
    -newkey rsa:2048
    -keyout /etc/ssl/private/selfsigned.key
    -out /etc/ssl/certs/selfsigned.crt
    -subj "/CN={{ server_public_ip }}"
    -addext "subjectAltName=IP:{{ server_public_ip }}"
  args:
    creates: /etc/ssl/certs/selfsigned.crt
```

9. **Role: webapp**

task13_webapp_defaults_main.png — content of roles/webapp/defaults/main.yml.

```yaml
#SPDX-License-Identifier: MIT-0
# defaults file for webapp
nginx_user: nginx
nginx_worker_processes: auto
nginx_worker_connections: 1024
nginx_error_log_level: notice

# Webapp settings
web_root: /usr/share/nginx/html
web_index_file: index.php
~
~
~
~
~
~
```

task13_webapp_files_index_php.png — content of roles/webapp/files/index.php.

```php
        }
        .label {
            font-weight: bold;
            color: #ffd700;
        }
        .info a {
            color: white;            /* same as other values */
            text-decoration: none;  /* remove underline */
            font-weight: normal;
        }

        .info a:hover {
            text-decoration: underline; /* optional: underline on hover */
        }
    </style>
</head>
<body>
    <div class="container">
        <h1>🚀 Nginx Front End Web Server </h1>

        <div class="info"><span class="label">Hostname:</span> <?= htmlspecialchars($hostname) ?></div>
        <div class="info"><span class="label">Instance ID:</span> <?= htmlspecialchars($instance_id) ?></div>
        <div class="info"><span class="label">Private IP:</span> <?= htmlspecialchars($private_ip) ?></div>
        <div class="info"><span class="label">Public IP:</span> <?= htmlspecialchars($public_ip) ?></div>
        <div class="info"><span class="label">Public DNS:</span>
            <a href="https://<?= htmlspecialchars($public_dns) ?>" target="_blank">
            https://<?= htmlspecialchars($public_dns) ?></a>
        </div>
        <div class="info"><span class="label">Deployed:</span> <?= $deployed_date ?></div>
        <div class="info"><span class="label">Status:</span> ✅ Active and Running</div>
        <div class="info"><span class="label">Managed By:</span> Terraform + Ansible</div>
    </div>
</body>
</html>
```

task13_webapp_handlers_main.png — content of roles/webapp/handlers/main.yml.

```
#SPDX-License-Identifier: MIT-0
# handlers file for webapp

- name: Restart nginx
  service:
    name: nginx
    state: restarted

- name: Restart php-fpm
  service:
    name: php-fpm
    state: restarted
~
~
~
```

task13_webapp_templates_nginx_conf.png — content of roles/webapp/templates/nginx.conf.j2.

```
    }

    server {
        listen 443 ssl;
        server_name {{ server_public_ip }};

        ssl_certificate /etc/ssl/certs/selfsigned.crt;
        ssl_certificate_key /etc/ssl/private/selfsigned.key;

        location / {
            root {{ web_root }};
            index {{ web_index_file }} index.html index.htm;
    #       proxy_pass http://158.252.94.241:80;
    #       proxy_pass http://backend_servers;

            location / {
                try_files $uri $uri/ =404;
            }

            # 🔴 This block is necessary for Php Website
            location ~ \.php$ {
                include fastcgi_params;
                fastcgi_pass unix:/run/php-fpm/www.sock;
                fastcgi_index index.php;
                fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
            }
        }
    }

    server {
        listen 80;
        server_name _;
        return 301 https://$host$request_uri;
    }
}
-- INSERT --                                                        62,2
```

task13_webapp_tasks_main.png — content of roles/webapp/tasks/main.yml.

```yaml
#SPDX-License-Identifier: MIT-0
# tasks file for webapp

- name: Install PHP packages
  yum:
    name:
      - php-fpm
      - php-curl
    state: present
  notify: Restart php-fpm

- name: Copy PHP website
  copy:
    src: index.php
    dest: "{{ web_root }}/{{ web_index_file }}"
    owner: nginx
    group: nginx
    mode: '0644'
  notify: Restart nginx

- name: Deploy nginx config
  template:
    src: nginx.conf.j2
    dest: /etc/nginx/nginx.conf
  notify: Restart nginx

- name: Start and enable php-fpm
  service:
    name: php-fpm
    state: started
    enabled: true
```

10. Final role-based playbook ansible/my-playbook.yaml:

task13_my_playbook_roles.png — final my-playbook.yaml referencing three roles.

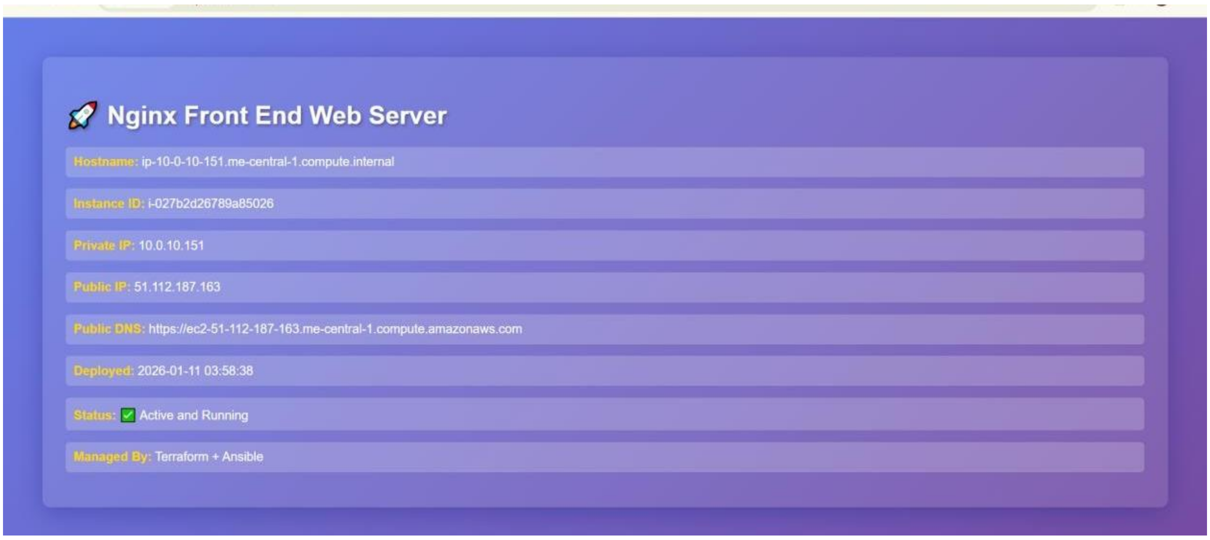
```yaml
---
- name: Deploy NGINX Web Stack with SSL and PHP
  hosts: nginx
  become: true
  roles:
    - nginx
    - ssl
    - webapp
~
~
~
~
```

11. Run:

task13_ansible_play_roles.png — playbook output with all roles executed.

```
@23-22411-061-rgb ➜ /workspaces/terraform_machine/ansible (main) $ ansible-playbook -i inventory/hosts my-playbook.yam
l
TASK [ssl : Get IMDSv2 token] ***************************************************************************
ok: [51.112.187.163]

TASK [ssl : Get public IP] ******************************************************************************
ok: [51.112.187.163]

TASK [ssl : Save public IP as fact] *********************************************************************
ok: [51.112.187.163]

TASK [ssl : Generate self-signed certificate] ***********************************************************
changed: [51.112.187.163]

TASK [webapp : Install PHP packages] ********************************************************************
changed: [51.112.187.163]

TASK [webapp : Copy PHP website] ************************************************************************
changed: [51.112.187.163]

TASK [webapp : Deploy nginx config] *********************************************************************
changed: [51.112.187.163]

TASK [webapp : Start and enable php-fpm] ****************************************************************
changed: [51.112.187.163]

RUNNING HANDLER [webapp : Restart nginx] ****************************************************************
changed: [51.112.187.163]

RUNNING HANDLER [webapp : Restart php-fpm] **************************************************************
changed: [51.112.187.163]

PLAY RECAP **********************************************************************************************
51.112.187.163             : ok=16   changed=9    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

@23-22411-061-rgb ➜ /workspaces/terraform_machine/ansible (main) $
```

Task13_php_https_browser_roles.png — browser showing final PHP page via HTTPS.



Cleanup

1. From the Terraform root:

cleanup_terraform_destroy.png — terminal showing destroy output.

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                                   bash
@23-22411-057-sudo →/workspaces/terraform_machine (main) $ terraform destroy -auto-approve
          - "Name" = "dev-default-sg"
        } -> null
      - tags_all               = {
          - "Name" = "dev-default-sg"
        } -> null
      - vpc_id                 = "vpc-0ade7c3c4a67a6421" -> null
        # (1 unchanged attribute hidden)
    }

Plan: 0 to add, 0 to change, 7 to destroy.

Changes to Outputs:
  - webserver_public_ips = [
      - "158.252.83.114",
    ] -> null
module.myapp-subnet.aws_default_route_table.main_rt: Destroying... [id=rtb-0b063f601309188d0]
module.myapp-webserver[0].aws_instance.myapp-server: Destroying... [id=i-0782572662a34a57c]
module.myapp-subnet.aws_default_route_table.main_rt: Destruction complete after 0s
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destroying... [id=igw-0cf955965564dac4c]
module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0782572662a34a57c, 00m10s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-0cf955965564dac4c, 00m10s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destruction complete after 17s
module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0782572662a34a57c, 00m20s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Destruction complete after 20s
module.myapp-webserver[0].aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-0]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destroying... [id=subnet-0ccedd3e703d0f611]
module.myapp-webserver[0].aws_security_group.web_sg: Destroying... [id=sg-05d304624bf559a84]
module.myapp-webserver[0].aws_key_pair.ssh-key: Destruction complete after 1s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 1s
module.myapp-webserver[0].aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0ade7c3c4a67a6421]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 7 destroyed.
@23-22411-057-sudo →/workspaces/terraform_machine (main) $
```

2. Verify state:

cleanup_tfstate.png — terraform.tfstate showing no active resources or is empty.

```
● @23-22411-057-sudo →/workspaces/terraform_machine/ansible/roles (main) $ cat terraform.tfstate
  {
    "version": 4,
    "terraform_version": "1.14.3",
    "serial": 1,
    "lineage": "09b172c8-0e22-2538-46c5-44c42ab2f7c8",
    "outputs": {},
    "resources": [],
    "check_results": null
  }
○ @23-22411-057-sudo →/workspaces/terraform_machine/ansible/roles (main) $
```

3. Confirm that no EC2 instances remain in AWS console.

cleanup_aws_console.png — AWS EC2 console showing zero running instances for this lab.